ASYMPTOTIC SHIFT THRESHOLDS FOR PÓLYA–SCHUR MULTIPLIERS AND A CERTIFIED LOWER BOUND ON THE DE BRUIJN–NEWMAN CONSTANT OF THE DAVENPORT–HEILBRONN FUNCTION

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Abstract. We establish sufficient asymptotic shift conditions for real entire Pólya–Schur multipliers $G(-\partial_z^2)$ acting on the completed Davenport–Heilbronn function $\Xi_{\mathrm{DH}}(z)$ and certify unconditional bounds on its de Bruijn–Newman constant Λ_{DH}. Via the Stieltjes logarithmic derivative $S_G(u)$ relative to $\sigma^* := \sup\{\operatorname{Re}\rho : \mathcal{D}_{\mathrm{DH}}(\rho) = 0\}$, we establish persistence versus finiteness thresholds for non-real zeros and construct tower-lacunary canonical products of every upper order $\rho \in (1/2, 1)$ preserving infinitely many off-line zeros. For Pólya's sinc multiplier, singularity-free interval continuation across $h \in [0, 5/8]$ and a doubling identity yield $\sigma^* - 1/2 = h^*_{\mathrm{global}}(G_{\cosh}; \Xi_{\mathrm{DH}}) < 5/8 < h^*_{\mathrm{global}}(G_{\mathrm{sinc}}; \Xi_{\mathrm{DH}}) \le 2\sigma^* - 1$, while conditionally under RH and GRH for $L(s, \chi_{5,\mathrm{quad}})$ only finitely many (≥ 8) non-real zeros remain at $h_* = \sigma^* - 1/2$. Finally, Interval–Newton enclosures and the Ki–Kim–Lee method yield $28825920237/125000000000 < \Lambda_{\mathrm{DH}} < 2(\sigma^* - 1/2)^2$.

1. Introduction

Let χ_5 denote the odd primitive Dirichlet character modulo 5 given by $\chi_5(1) = 1$, $\chi_5(2) = \mathrm{i}$, $\chi_5(3) = -\mathrm{i}$, $\chi_5(4) = -1$, and $\chi_5(5) = 0$, and consider the self-dual Dirichlet combination introduced by Davenport and Heilbronn [8] (see Titchmarsh [29, §10.25]):

$$\kappa := \frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1}, \qquad \mathcal{D}_{\mathrm{DH}}(s) := \frac{1 - \mathrm{i}\kappa}{2} L(s, \chi_5) + \frac{1 + \mathrm{i}\kappa}{2} L(s, \overline{\chi_5}) = \sum_{n=1}^{\infty} a_{\mathrm{DH}}(n)\, n^{-s}. \tag{1.1}$$

In the canonical normalization of Riemann [25], Pólya [22, 23], de Bruijn [9], and Newman [20], the completed entire function and its theta-kernel cosine representation take the form

$$\begin{aligned} \xi_{\mathrm{DH}}(s) &:= \left(\frac{5}{\pi}\right)^{(s+1)/2} \Gamma\left(\frac{s+1}{2}\right) \mathcal{D}_{\mathrm{DH}}(s), \\ \Xi_{\mathrm{DH}}(z) &:= \xi_{\mathrm{DH}}\left(\tfrac{1}{2} + \tfrac{\mathrm{i}z}{2}\right) = 2\int_0^{\infty} \Phi_{\mathrm{DH}}(u) \cos(zu)\, \mathrm{d}u, \end{aligned} \tag{1.2}$$

where $\Phi_{\mathrm{DH}}(u) := 4\mathrm{e}^{3u} \sum_{n=1}^{\infty} n\, a_{\mathrm{DH}}(n)\, \mathrm{e}^{-\pi n^2 \mathrm{e}^{4u}/5}$ satisfies $\Phi_{\mathrm{DH}}(-u) = \Phi_{\mathrm{DH}}(u)$, encoding $\xi_{\mathrm{DH}}(s) = \xi_{\mathrm{DH}}(1 - s)$ and $\Xi_{\mathrm{DH}}(z) = \Xi_{\mathrm{DH}}(-z) = \overline{\Xi_{\mathrm{DH}}(\bar{z})}$. Whereas Riemann's $\Xi(z)$ links critical-line zero geometry to classical criteria of Csordas, Smith, and Varga [7] and satisfies $0 \le \Lambda_\zeta \le 1/5$ [27, 24, 21], $\mathcal{D}_{\mathrm{DH}}(s)$ lacks an Euler product and possesses infinitely many zeros both in $\operatorname{Re}(s) > 1$ and off the critical line inside $0 < \operatorname{Re}(s) < 1$ [8, 28, 2, 5]. Bombieri and Ghosh [5] proved that $\sigma^* := \sup\{\operatorname{Re}\rho : \mathcal{D}_{\mathrm{DH}}(\rho) = 0\}$ is determined via Kronecker's theorem by $\sum_{p \equiv 2,3 \pmod 5} \arctan(p^{-\sigma^*}) = \frac{\pi}{2} - \arctan\kappa$; at the same time, Righetti [26] showed that the real parts of zeros of general linear combinations of Euler products in $\operatorname{Re}(s) > 1$ need not be dense in $(1, \sigma^*)$.

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Two developments in the de Bruijn–Newman literature directly motivate the analytical framework of the present paper. First, Dobner [10] proved Newman's conjecture for the extended Selberg class by approximating the heat-evolved L-function $\xi_t(s)$ via the modified Dirichlet series $\sum_{n=1}^{\infty} a(n)\,\mathrm{e}^{(t/4)\log^2 n} n^{-s}$; in Section 2 we generalize Dobner's Gaussian multiplier $G_t(w) = \mathrm{e}^{tw}$ to real entire Pólya–Schur multipliers $G(w) = \mathrm{e}^{\lambda w}\prod_{j=1}^{\infty}(1 + w/w_j)$, replacing $\mathrm{e}^{(t/4)\log^2 n}$ by the canonical-product ratio $h_G(\Omega(z) - \log n)/h_G(\Omega(z))$ governed by the Stieltjes logarithmic derivative $S_G(u) = \frac{\mathrm{d}}{\mathrm{d}u}\log G(u^2)$. Second, Ki, Kim, and Lee [14] established the strict upper bound $\Lambda_\zeta < 1/2$ (recently refined for Dirichlet L-functions in [11]) by proving asymptotic simplicity of zeros of H_t at large height and applying de Bruijn's hyperbola inequality on the complementary compact box; in Section 4 we adapt the Ki–Kim–Lee argument to Ξ_{DH} to make the upper bound $\Lambda_{\mathrm{DH}} < 2(\sigma^* - \frac{1}{2})^2$ explicit alongside our certified interval-arithmetic lower bound.

Theorem 1.1. *Let $\mathcal{G}_0$ denote the class of real entire Pólya–Schur multipliers $G(w) = \mathrm{e}^{\lambda w}\prod_{j=1}^{\infty}(1 + w/w_j)$ with $\lambda \geq 0$, $0 < w_1 \leq w_2 \leq \cdots$, and $\sum_{j=1}^{\infty} w_j^{-1} < \infty$, acting via $G(-\partial_z^2)\Xi_{\mathrm{DH}}(z) = 2\int_0^{\infty} G(u^2)\Phi_{\mathrm{DH}}(u)\cos(zu)\,\mathrm{d}u$. Define the Stieltjes logarithmic derivative $S_G(u) := \frac{\mathrm{d}}{\mathrm{d}u}\log G(u^2) = 4\lambda u + \sum_{j=1}^{\infty}\frac{2u}{w_j+u^2}$ $(u > 0)$, and let $N_{\mathrm{off}}(F;T)$ count the non-real zeros of $F(z)$ in $|\operatorname{Re} z| \leq 2T$ with multiplicity.*

(i) Sufficient Stieltjes shift conditions and lacunary obstruction.
   (a) *If $\lambda = 0$ and $\sum_{j=1}^{\infty} w_j^{-1/2} < \infty$ (including every polynomial and every multiplier of upper order $\rho(G) < 1/2$), then $\lim_{u\to\infty} S_G(u) = 0$ and $N_{\mathrm{off}}(G(-\partial_z^2)\Xi_{\mathrm{DH}};T) \gg_G T$ inside $|\operatorname{Im} z| > 1$. In particular, $(1-\beta\partial_z^2)\Xi_{\mathrm{DH}}(z)$ has infinitely many non-real zeros for every $\beta > 0$.*
   (b) *If $\lambda = 0$ and $\lim_{u\to\infty} S_G(u) = \tau \in (0,\infty)$, then $N_{\mathrm{off}}(G(-\partial_z^2)\Xi_{\mathrm{DH}};T) \gg_G T$ for every $0 < \tau < 2\sigma^* - 1$, whereas $N_{\mathrm{off}}(G(-\partial_z^2)\Xi_{\mathrm{DH}};\infty) < \infty$ for every $\tau > 2\sigma^* - 1$.*
   (c) *Every $G \in \mathcal{G}_0$ with $\lambda > 0$ or lower order $\underline{\rho}(G) > 1/2$ satisfies $\lim_{u\to\infty} S_G(u) = +\infty$. Moreover, $N_{\mathrm{off}}(G(-\partial_z^2)\Xi_{\mathrm{DH}};\infty) < \infty$ whenever $\lambda > 0$, or whenever $\lambda = 0$ and $w_{j+1} - w_j = o(\sqrt{w_j})$ as $j \to \infty$ (in particular for $w_j \sim c\,j^{1/\rho}$ with $\rho \in (1/2,1)$). Conversely, for every $\rho \in (1/2,1)$ there exists a lacunary canonical product $G \in \mathcal{G}_0$ of upper order $\rho(G) = \rho$ and lower order $\underline{\rho}(G) = 0$ such that $\liminf_{u\to\infty} S_G(u) = 0$ and $G(-\partial_z^2)\Xi_{\mathrm{DH}}(z)$ has infinitely many non-real zeros in $|\operatorname{Im} z| > 1$.*

(ii) Strict separation between asymptotic shift and global real-rootedness. *Comparing $G_{\cosh,h}(w) := \cosh(2h\sqrt{w})$ and Pólya's hyperbolic sinc multiplier $G_{\mathrm{sinc},h}(w) := \frac{\sinh(2h\sqrt{w})}{2h\sqrt{w}}$, both having $\lim_{u\to\infty} S_G(u) = 2h$, the global thresholds $h^*_{\mathrm{global}}(G_\bullet;\Xi_{\mathrm{DH}}) := \inf\{h > 0 : G_{\bullet,h}(-\partial_z^2)\Xi_{\mathrm{DH}}$ has only real zeros$\}$ satisfy*

$$h^*_{\mathrm{global}}(G_{\cosh};\Xi_{\mathrm{DH}}) = \sigma^* - \tfrac{1}{2} < \frac{5}{8} < h^*_{\mathrm{global}}(G_{\mathrm{sinc}};\Xi_{\mathrm{DH}}) \leq 2\sigma^* - 1. \tag{1.3}$$

(iii) Unconditional bounds on Λ_{DH}. *Under backward heat evolution $H_t^{\mathrm{DH}}(z) := \mathrm{e}^{-t\partial_z^2}\Xi_{\mathrm{DH}}(z)$, the de Bruijn–Newman constant $\Lambda_{\mathrm{DH}} := \inf\{t \in \mathbb{R} : H_t^{\mathrm{DH}}$ has only real zeros$\}$ satisfies*

$$\frac{28825920237}{125000000000} < \Lambda_{\mathrm{DH}} < 2\Big(\sigma^* - \tfrac{1}{2}\Big)^2. \tag{1.4}$$

The constants in Theorem 1.1 are certified in interval ball arithmetic (Lemma 3.1) to satisfy

$$\begin{aligned} \sigma^* &= 1.120362498\ldots, & h^*_\infty &:= \sigma^* - \tfrac{1}{2} = 0.620362498\ldots, \\ 2\sigma^* - 1 &= 1.240724996\ldots, & 2(h^*_\infty)^2 &= 0.769699258\ldots \end{aligned} \tag{1.5}$$

For comparison, Platt and Trudgian [21] proved unconditionally that $\Lambda_\zeta \leq \frac{1}{5} = 0.2$, so (1.4) implies in particular $\Lambda_{\mathrm{DH}} > \frac{28825920237}{125000000000} = 0.230607361896 > \Lambda_\zeta$.

1.1. **Proof overview.** The proof of Theorem 1.1 rests on three technical components:

(1) *Conjugate-diagram Borel–Laplace asymptotics and single-zero syndetic translation (Section 2):* When $\lambda = 0$ and $\lim_{u\to\infty} S_G(u) = \tau_0 \in [0,\infty)$, Pólya's conjugate-diagram representation transfers $G(-\partial_z^2)\Xi_{\mathrm{DH}}(z)$ to a compact stadium contour around $[-\frac{\tau_0}{2}, \frac{\tau_0}{2}]$, generalizing Dobner's modified Dirichlet series [10] via $h_G(\Omega(z) - \log n)/h_G(\Omega(z))$. For $0 \le \tau_0 < 2\sigma^* - 1$, translating a single off-line zero in $\mathrm{Re}(s) > 1$ by Dirichlet simultaneous approximation avoids Righetti's non-density phenomenon [26] and yields $\gg_G T$ non-real zeros via Rouché's theorem; for $\tau_0 > 2\sigma^* - 1$, diagonal Bohr compactness on $\prod_p S^1$ and seamless Rouché rectangles along $\mathbb{R}$ exclude non-real zeros above a finite height, while subproduct extraction on disjoint intervals (t_m, t_{m+1}) extends finiteness to $w_{j+1} - w_j = o(\sqrt{w_j})$.

(2) *Singularity-free interval continuation across $[0, 5/8]$, doubling upper bound, and conditional boundary resolution for $G_{\mathrm{sinc},h}$ (Section 3):* Whereas $G_{\cosh,h}(-\partial_z^2)\Xi_{\mathrm{DH}}$ becomes globally real-rooted at $h = \sigma^* - \frac{1}{2}$, a 384-bit, 360-ball singularity-free Interval–Newton continuation from ρ_{44838} at $h = 0$ across all $h \in [0, \frac{5}{8}]$ (and to $h = \frac{625121}{10^6} = 0.625121$) certifies $h^*_{\mathrm{global}}(G_{\mathrm{sinc}}; \Xi_{\mathrm{DH}}) > \frac{5}{8}$ independently of Section 2, while the interval-splitting doubling identity $G_{\mathrm{sinc},2h_1}(-\partial_z^2)\Xi_{\mathrm{DH}} = G_{\cosh,h_1}(-\partial_z^2)G_{\mathrm{sinc},h_1}(-\partial_z^2)\Xi_{\mathrm{DH}}$ immediately yields $h^*_{\mathrm{global}}(G_{\mathrm{sinc}}; \Xi_{\mathrm{DH}}) \le 2\sigma^* - 1$. At the critical boundary $h_* = \sigma^* - \frac{1}{2}$ ($\tau = 2\sigma^* - 1$), we also establish conditionally on RH for $\zeta(s)$ and GRH for $L(s, \chi_{5,\mathrm{quad}})$ that $8 \le N_{\mathrm{off}}(G_{\mathrm{sinc},h_*}(-\partial_z^2)\Xi_{\mathrm{DH}}; \infty) < \infty$ (Proposition 3.3).

(3) *Ki–Kim–Lee asymptotic simplicity and 512-bit Interval–Newton heat enclosures (Sections 4–5):* Writing $H_t^{\mathrm{DH}}(2\gamma - 2\mathrm{i}\delta) = A_t(\gamma,\delta) + \overline{A_t(\gamma,-\delta)}$ on $[0,\infty)$ and shifting the Gaussian contour by $\mathrm{i}\frac{\pi t}{8}$ yields an explicit height $\gamma_0(t)$ above which all zeros of H_t^{DH} are real and simple, from which the Ki–Kim–Lee hyperbola argument [14, 9] gives Corollary 4.4, while a 512-bit `FLINT/Arb` Interval–Newton contraction with third-order derivative Taylor enclosure near ρ_{240} at $t = \frac{28825920237}{125000000000} = 0.230607361896$ certifies $\Lambda_{\mathrm{DH}} > \frac{28825920237}{125000000000}$.

Combining Component (1) with Proposition 2.7 establishes Theorem 1.1(i); Component (2) proves (1.3) independently of Theorem 1.1(i); and Component (3) proves (1.4).

1.2. **Notation and conventions.** Throughout, $s = \sigma + \mathrm{i}\gamma$ ($\sigma, \gamma \in \mathbb{R}$) denotes the Dirichlet variable and $z = x - 2\mathrm{i}\delta = 2\gamma - 2\mathrm{i}\delta$ ($\delta = \sigma - \frac{1}{2}$) denotes the spectral variable in (1.2), mapping $\sigma = \frac{1}{2}$ to $z \in \mathbb{R}$ and $|\sigma - \frac{1}{2}| \le h^*_\infty$ to $|\mathrm{Im}\, z| \le 2h^*_\infty$. Under $\partial_t H_t = -\partial_z^2 H_t$, Λ matches the convention of de Bruijn [9], Newman [20], Rodgers–Tao [27], Polymath [24], and Platt–Trudgian [21]; in the rescaled variable $w = z/2$ with $\widetilde{\Xi}(w) = \xi(\frac{1}{2} + \mathrm{i}w)$, one has $\widetilde{\Lambda} = \Lambda/4$. We write $\Gamma_{\mathrm{DH}}(s) := (5/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})$. Estimates $f \ll_\alpha g$ or $f = \mathcal{O}_\alpha(g)$ denote $|f| \le C(\alpha)|g|$ with $C(\alpha)$ depending only on α. Closed interval balls $[m \pm r]$ denote rigorous enclosures $\{w : |w - m| \le r\}$ computed in `FLINT/Arb` [12].

2. Sufficient Stieltjes Shift Conditions and Root-Gap Obstruction

Let $G(w) = \mathrm{e}^{\lambda w}\prod_{j=1}^\infty (1 + w/w_j) \in \mathcal{G}_0$, where $\lambda \ge 0$, $0 < w_1 \le w_2 \le \cdots$, and $\sum_{j=1}^\infty w_j^{-1} < \infty$. On every horizontal strip $|\mathrm{Im}\, u| \le \frac{\pi}{8} - \delta_0$ ($\delta_0 \in (0, \frac{\pi}{8})$), the elementary bound $1 + r \le \mathrm{e}^r$ gives $|G(u^2)| \le \mathrm{e}^{(\lambda + \sum_{j=1}^\infty w_j^{-1})|u|^2}$, whereas $\mathrm{Re}(\mathrm{e}^{4u}) = \mathrm{e}^{4\,\mathrm{Re}\,u}\cos(4\,\mathrm{Im}\,u) \ge \sin(4\delta_0)\,\mathrm{e}^{4\,\mathrm{Re}\,u}$ implies $|\Phi_{\mathrm{DH}}(u)| \ll_{\delta_0} \mathrm{e}^{3|\mathrm{Re}\,u| - \frac{\pi}{5}\sin(4\delta_0)\mathrm{e}^{4|\mathrm{Re}\,u|}}$. Consequently, the integral

$$G(-\partial_z^2)\Xi_{\mathrm{DH}}(z) = 2\int_0^\infty G(u^2)\,\Phi_{\mathrm{DH}}(u)\cos(zu)\,\mathrm{d}u = \int_{-\infty}^\infty G(u^2)\,\Phi_{\mathrm{DH}}(u)\,\mathrm{e}^{\mathrm{i}zu}\,\mathrm{d}u \tag{2.1}$$

converges absolutely and locally uniformly on $\mathbb{C}$ to a real-even entire function of order 1. Because $G(u^2) > 0$ for all $u \in \mathbb{R}$ and Φ_{DH} is continuous and not identically zero, Fourier uniqueness ensures $G(-\partial_z^2)\Xi_{\mathrm{DH}} \not\equiv 0$. Furthermore, for any factorization $G = G^{(1)}G^{(2)}$ in $\mathcal{G}_0$, the finite truncations $G_N^{(2)}(w) := \mathrm{e}^{\lambda_2 w}\prod_{j=1}^N (1 + w/w_{2,j})$ satisfy $0 < G_N^{(2)}(u^2) \le G^{(2)}(u^2)$ on

$\mathbb{R}$, so dominated convergence in (2.1) yields $G_N^{(2)}(-\partial_z^2)[G^{(1)}(-\partial_z^2)\Xi_{\mathrm{DH}}](z) \to G(-\partial_z^2)\Xi_{\mathrm{DH}}(z)$ locally uniformly on $\mathbb{C}$.

We begin by recording two classical zero-preservation lemmas for Laguerre–Pólya differential operators.

Lemma 2.1 (Pólya [22, 23], Benz [3]; cf. Lagarias–Suzuki [16])**.** *Let $F(z) \not\equiv 0$ be a real-even entire function of genus 0 or 1 whose zeros all satisfy $|\operatorname{Im} z| \le 2h_0$ ($h_0 > 0$).*

(i) *For every $G \in \mathcal{G}_0$ such that $G_N(-\partial_z^2)F \to G(-\partial_z^2)F \not\equiv 0$ locally uniformly on $\mathbb{C}$, all zeros of $G(-\partial_z^2)F(z)$ lie in the strip $|\operatorname{Im} z| \le 2h_0$.*

(ii) *For $G_{\cosh,h}(w) = \cosh(2h\sqrt{w})$, all zeros of $G_{\cosh,h}(-\partial_z^2)F(z) = \frac{1}{2}[F(z-2\mathrm{i}h)+F(z+2\mathrm{i}h)]$ are real whenever $h \ge h_0$, and real and simple whenever $h > h_0$.*

Proof. For (i), let $H(z) = c\, z^m \mathrm{e}^{az} \prod_n (1 - z/\zeta_n)\mathrm{e}^{\mu_n z/\zeta_n}$ ($c \in \mathbb{R}\setminus\{0\}$, $a \in \mathbb{R}$, $\mu_n \in \{0,1\}$, $\sum_{\zeta_n \ne 0} |\zeta_n|^{-2} < \infty$) be any real entire function of genus 0 or 1 whose zeros satisfy $|\operatorname{Im}\zeta_n| \le 2h_0$. Grouping non-real zeros into conjugate pairs $\{\zeta_n, \overline{\zeta_n}\}$ cancels $\operatorname{Im}(\mu_n/\zeta_n + \mu_n/\overline{\zeta_n}) = 0$, so taking imaginary parts of $H'(z_0)/H(z_0)$ gives the absolutely convergent sum $\operatorname{Im}(H'(z_0)/H(z_0)) = -\frac{m \operatorname{Im} z_0}{|z_0|^2} - \sum_n \frac{\operatorname{Im} z_0 - \operatorname{Im}\zeta_n}{|z_0-\zeta_n|^2} < 0$ for every $\operatorname{Im} z_0 > 2h_0$, whence $(1 \pm \alpha\partial_z)H(z_0) \ne 0$ for all $\alpha \in \mathbb{R}$. Iterating across the linear factors of $G_N(-\partial_z^2)F$ (with de Bruijn [9, Thm. 13] for $\mathrm{e}^{-\lambda\partial_z^2}$) and passing to the limit via Hurwitz's theorem gives (i). For (ii), all zeros of $\omega(z) := F(z + 2\mathrm{i}h)$ lie in $\operatorname{Im} z \le 2(h_0 - h) \le 0$, so factorwise comparison gives $|\omega(z)| > |\overline{\omega(\bar z)}| = |F(z - 2\mathrm{i}h)|$ on $\operatorname{Im} z > 0$, and the Hermite–Biehler theorem yields simplicity on $\mathbb{R}$ when $h > h_0$. □

Lemma 2.2 (Non-real zero monotonicity; cf. Pólya [23], Craven–Csordas–Smith [6])**.** *Let $F(z)$ be a real entire function of order at most 1 with $N_{\mathrm{off}}(F;\infty) = 2M < \infty$. Then for every $G \in \mathcal{G}_0$ such that $G_N(-\partial_z^2)F \to G(-\partial_z^2)F \not\equiv 0$ locally uniformly on $\mathbb{C}$, one has $N_{\mathrm{off}}(G(-\partial_z^2)F;\infty) \le 2M$.*

Proof. Factor $F = P_{2M}E$, where $\deg P_{2M} = 2M$ and E lies in the Laguerre–Pólya class of order at most 1. Approximating E locally uniformly by real-rooted polynomials Q_K of degree K, the polynomial $R_K := P_{2M}Q_K$ of degree $2M + K$ has at least K real roots counted with multiplicity. For any $\alpha \in \mathbb{R}\setminus\{0\}$, Rolle's theorem applied to $\mathrm{e}^{x/\alpha}R_K(x)$ between consecutive real roots and along the half-line where $\mathrm{e}^{x/\alpha}R_K(x) \to 0$ produces at least K real roots of $(1 + \alpha\partial_z)R_K$, leaving at most $2M$ non-real roots. Letting $K \to \infty$ and $G_N(-\partial_z^2)F \to G(-\partial_z^2)F$ via Hurwitz's theorem preserves $N_{\mathrm{off}} \le 2M$. □

Remark 2.3. For the single derivative $F \mapsto F'$, the refined asymptotic $N_{\mathrm{off}}(F;T) - N_{\mathrm{off}}(F';T) = 2D(F;T) + \mathcal{O}(\log T)$ counting real critical points of F/F' was proved for Riemann's Ξ by Levinson and Montgomery [18] and for real entire functions of genus 0 or 1 by Ki and Kim [13]; here only the global finiteness bound of Lemma 2.2 is required.

We next isolate the vertical-strip growth, single-zero Diophantine translation, and supercritical lower-bound properties of $\mathcal{D}_{\mathrm{DH}}(s)$ and $\xi_{\mathrm{DH}}(s)$ needed in the contour analysis.

Lemma 2.4 (Vertical-strip convexity, syndetic zero translation, and supercritical lower bound)**.**

(i) *For every $\eta \in (0, \frac{1}{2})$ and $A \ge 2$, there exists a constant $C_0(\eta, A) > 0$ such that for all $\sigma \in [-A, A]$ and all $\tau \ge 2$,*

$$\left|\xi_{\mathrm{DH}}(\sigma + \mathrm{i}\tau)\right| \le C_0(\eta, A)\, \tau^{\frac{1}{2}\max(\sigma, 1-\sigma, 1+\eta)}\, \mathrm{e}^{-\frac{\pi}{4}\tau}. \tag{2.2}$$

(ii) *For every $\sigma_{\mathrm{low}} \in [1, \sigma^*)$, there exist a zero $\rho_* = \beta_0 + \mathrm{i}\gamma_*$ of $\mathcal{D}_{\mathrm{DH}}(s)$ with $\beta_0 \in (\sigma_{\mathrm{low}}, \sigma^*]$, radii $\eta_0 := \frac{1}{4}(\beta_0 - \sigma_{\mathrm{low}}) > 0$ and $r_0 \in (0, \min(\eta_0/2, 1/4))$, a constant $\mu_0 > 0$, and a threshold $T_0 \ge 4$ such that for every $T \ge T_0$ there exist $M(T) \gg_{\sigma_{\mathrm{low}}} T$ real shifts $\tau_1 < \tau_2 < \cdots < \tau_{M(T)}$ in $[T, 2T]$ satisfying $\tau_{m+1} - \tau_m \ge 1$ such that on every translated*

circle $|w-(\rho_*+\mathrm{i}\tau_m)|=r_0$,

$$|\mathcal{D}_{\mathrm{DH}}(w)-\mathcal{D}_{\mathrm{DH}}(w-\mathrm{i}\tau_m)|<\frac{\mu_0}{2},\qquad |\mathcal{D}_{\mathrm{DH}}(w)|>\frac{\mu_0}{2},\tag{2.3}$$

and $\mathcal{D}_{\mathrm{DH}}(w)$ *has at least one zero inside each disjoint disk* $|w-(\rho_*+\mathrm{i}\tau_m)|<r_0$.

(iii) *For every* $\varepsilon>0$, *there exists a constant* $c(\varepsilon)>0$ *such that* $|\mathcal{D}_{\mathrm{DH}}(w)|\ge c(\varepsilon)$ *for all* $w\in\mathbb{C}$ *with* $\operatorname{Re} w\ge\sigma^*+\varepsilon$.

Proof. (i) For $\sigma\in[1+\eta,A]$ and $\tau\ge 2$, $|\mathcal{D}_{\mathrm{DH}}(\sigma+\mathrm{i}\tau)|\le\zeta(1+\eta)$, while Stirling's formula gives $|\Gamma_{\mathrm{DH}}(\sigma+\mathrm{i}\tau)|\ll_A \tau^{\sigma/2}\mathrm{e}^{-\frac{\pi}{4}\tau}$, establishing (2.2) on $[1+\eta,A]$ since $\max(\sigma,1-\sigma,1+\eta)=\sigma$. By the functional equation $|\xi_{\mathrm{DH}}(\sigma+\mathrm{i}\tau)|=|\xi_{\mathrm{DH}}(1-\sigma-\mathrm{i}\tau)|=|\xi_{\mathrm{DH}}(1-\sigma+\mathrm{i}\tau)|$, replacing A by $A+1$ yields $|\xi_{\mathrm{DH}}(\sigma+\mathrm{i}\tau)|\ll_{\eta,A}\tau^{(1-\sigma)/2}\mathrm{e}^{-\frac{\pi}{4}\tau}$ for $\sigma\in[-A,-\eta]$, where $\max(\sigma,1-\sigma,1+\eta)=1-\sigma$. Finally, on the strip $\sigma\in[-\eta,1+\eta]$, $\mathcal{D}_{\mathrm{DH}}(w)$ has finite order, satisfies $|\mathcal{D}_{\mathrm{DH}}(1+\eta+\mathrm{i}\tau)|\le\zeta(1+\eta)$ and $|\mathcal{D}_{\mathrm{DH}}(-\eta+\mathrm{i}\tau)|=|\Gamma_{\mathrm{DH}}(1+\eta-\mathrm{i}\tau)/\Gamma_{\mathrm{DH}}(-\eta+\mathrm{i}\tau)|\,|\mathcal{D}_{\mathrm{DH}}(1+\eta-\mathrm{i}\tau)|\ll_\eta\tau^{\frac12+\eta}$ for $\tau\ge2$, so the Phragmén–Lindelöf principle gives $|\mathcal{D}_{\mathrm{DH}}(\sigma+\mathrm{i}\tau)|\ll_\eta\tau^{\frac12(1+\eta-\sigma)}$; multiplying by $|\Gamma_{\mathrm{DH}}(\sigma+\mathrm{i}\tau)|\ll_\eta\tau^{\sigma/2}\mathrm{e}^{-\frac{\pi}{4}\tau}$ yields exponent $\frac12(1+\eta)=\frac12\max(\sigma,1-\sigma,1+\eta)$.

(ii) Because Righetti [26, Thm. 1.2] showed that the real parts of zeros of general linear combinations of Euler products in $\operatorname{Re}(s)>1$ need not be dense in $(1,\sigma^*)$, one cannot prescribe an arbitrary abscissa in $(\sigma_{\mathrm{low}},\sigma^*)$. Instead, by the supremum definition $\sigma^*=\sup\{\operatorname{Re}\rho:\mathcal{D}_{\mathrm{DH}}(\rho)=0\}>\sigma_{\mathrm{low}}$, we fix a single existing zero $\rho_*=\beta_0+\mathrm{i}\gamma_*$ of $\mathcal{D}_{\mathrm{DH}}(s)$ with $\beta_0\in(\sigma_{\mathrm{low}},\sigma^*]$ (and multiplicity $m_*\ge1$). Set $\eta_0:=\frac14(\beta_0-\sigma_{\mathrm{low}})>0$, so $\beta_0\ge1+4\eta_0$. Since the zeros of $\mathcal{D}_{\mathrm{DH}}\not\equiv0$ are isolated, there exists $r_0\in(0,\min(\eta_0/2,1/4))$ such that $\mathcal{D}_{\mathrm{DH}}(w)\ne0$ for $0<|w-\rho_*|\le r_0$, and compactness gives $\mu_0:=\min_{|w-\rho_*|=r_0}|\mathcal{D}_{\mathrm{DH}}(w)|>0$. Every w_0 with $|w_0-\rho_*|=r_0$ satisfies $\operatorname{Re}w_0\ge\beta_0-r_0>1+3\eta_0$. Choose $N_0\ge2$ such that $\sum_{n>N_0}n^{-(1+3\eta_0)}<\mu_0/6$, and then choose $\delta_{\mathrm{D}}\in(0,\frac14)$ such that $2\pi\delta_{\mathrm{D}}\sum_{n=2}^{N_0}n^{-(1+3\eta_0)}<\mu_0/6$. Let $p_1<\cdots<p_r$ be the primes $\le N_0$, set $L_0:=\log_2N_0$ and $Q:=\lceil L_0/\delta_{\mathrm{D}}\rceil\ge2$, and partition the r-torus $(\mathbb{R}/\mathbb{Z})^r$ into $K:=(2Q)^r$ half-open subcubes $C_1,\dots,C_K$ of ℓ^∞-diameter $\frac{1}{2Q}$. For each index ν such that $\{u\ge0:u\alpha \bmod 1\in C_\nu\}\ne\emptyset$, where $\alpha:=(\frac{\log p_1}{2\pi},\dots,\frac{\log p_r}{2\pi})$, pick one witness $u_\nu\ge0$ and set $U_0:=1+\max_\nu u_\nu<\infty$. For any $X\ge U_0$, the point $X\alpha \bmod 1$ lies in some non-empty C_ν, so $\tau(X):=X-u_\nu\in[X-U_0,X]$ satisfies $\max_{1\le\ell\le r}\operatorname{dist}(\tau(X)\frac{\log p_\ell}{2\pi},\mathbb{Z})\le\frac{1}{2Q}$. Setting $X_m:=T+m(U_0+1)$ for $1\le m\le\lfloor\frac{T}{U_0+1}\rfloor$ ($T\ge T_0:=2U_0+4$) produces $M(T)\gg_{\sigma_{\mathrm{low}}}T$ shifts $\tau_m:=\tau(X_m)\in[T,2T]$ with $\tau_{m+1}-\tau_m\ge1$. For every $2\le n=\prod_{\ell=1}^r p_\ell^{\nu_\ell(n)}\le N_0$, $\sum_{\ell=1}^r\nu_\ell(n)\le L_0$ implies $\operatorname{dist}(\tau_m\frac{\log n}{2\pi},\mathbb{Z})\le\frac{L_0}{2Q}<\delta_{\mathrm{D}}$ and $|n^{-\mathrm{i}\tau_m}-1|\le2\pi\delta_{\mathrm{D}}$. Hence for $|w_0-\rho_*|=r_0$,

$$|\mathcal{D}_{\mathrm{DH}}(w_0+\mathrm{i}\tau_m)-\mathcal{D}_{\mathrm{DH}}(w_0)|\le\sum_{n=2}^{N_0}n^{-(1+3\eta_0)}|n^{-\mathrm{i}\tau_m}-1|+2\sum_{n>N_0}n^{-(1+3\eta_0)}<\frac{\mu_0}{6}+\frac{\mu_0}{3}=\frac{\mu_0}{2},$$

giving (2.3), and Rouché's theorem produces $m_*\ge1$ zeros inside $|w-(\rho_*+\mathrm{i}\tau_m)|<r_0$.

(iii) For $\operatorname{Re}w\ge3$, $|\mathcal{D}_{\mathrm{DH}}(w)|\ge1-\sum_{n=2}^\infty n^{-3}=2-\zeta(3)>\frac34$. Suppose toward a contradiction that $\mathcal{D}_{\mathrm{DH}}(\sigma_k+\mathrm{i}\tau_k)\to0$ for a sequence with $\sigma_k\in[\sigma^*+\varepsilon,3]$ and $\tau_k\in\mathbb{R}$. Passing to a diagonal subsequence, we have $\sigma_k\to\sigma_\infty\in[\sigma^*+\varepsilon,3]$ and $p^{-\mathrm{i}\tau_k}\to\omega(p)\in S^1$ for every prime p; extending ω multiplicatively to $\mathbb{N}$, dominated convergence on $\sum_{n=1}^\infty n^{-(\sigma^*+\varepsilon)}<\infty$ yields $\frac{1-\mathrm{i}\kappa}{2}L_\omega(\sigma_\infty,\chi_5)+\frac{1+\mathrm{i}\kappa}{2}L_\omega(\sigma_\infty,\overline{\chi_5})=\lim_{k\to\infty}\mathcal{D}_{\mathrm{DH}}(\sigma_k+\mathrm{i}\tau_k)=0$, where $L_\omega(s,\chi):=\prod_p(1-\chi(p)\omega(p)p^{-s})^{-1}\ne0$. Thus $L_\omega(\sigma_\infty,\overline{\chi_5})/L_\omega(\sigma_\infty,\chi_5)=-\frac{1-\mathrm{i}\kappa}{1+\mathrm{i}\kappa}=\mathrm{e}^{+2\mathrm{i}(\frac{\pi}{2}-\arctan\kappa)}$. In this Euler product ratio, primes $p\equiv1,4\pmod5$ have real $\chi_5(p)\in\{\pm1\}$ and cancel identically for every ω, while each $p\equiv2,3\pmod5$ contributes $\frac{1-u_p}{1+u_p}$ with $|u_p|=p^{-\sigma_\infty}$, whose

principal argument is bounded in modulus by $2\arctan(p^{-\sigma_\infty})$. Since $\sigma_\infty \geq \sigma^* + \varepsilon > \sigma^*$,

$$\left|\sum_{p\equiv 2,3 \pmod 5} \arg\frac{1-u_p}{1+u_p}\right| \leq \sum_{p\equiv 2,3 \pmod 5} 2\arctan(p^{-\sigma_\infty}) < \sum_{p\equiv 2,3 \pmod 5} 2\arctan(p^{-\sigma^*}) = 2\Big(\frac{\pi}{2} - \arctan\kappa\Big) < \pi,$$

which prevents equality modulo 2π, a contradiction. □

We now establish the quantitative Borel–Laplace cap formula generalizing Dobner's modified Dirichlet-series approximation [10, Lem. 2.3] from the Gaussian multiplier $G_t(w) = \mathrm{e}^{tw}$ to any $G \in \mathcal{G}_0$ whose Stieltjes derivative converges to a finite limit.

Lemma 2.5 (Quantitative Borel–Laplace cap approximation; cf. Dobner [10])**.** *Let $G \in \mathcal{G}_0$ satisfy $\lambda = 0$ and $\lim_{u\to\infty} S_G(u) = \tau_0 \in [0,\infty)$, and define the even entire function $h_G(w) := G(\frac{1}{4}w^2) = \prod_{j=1}^\infty (1 + \frac{w^2}{4w_j})$.*

(i) *As $|w| \to \infty$ within the sector $|\arg w| \leq \frac{\pi}{3}$, one has $\frac{h_G'(w)}{h_G(w)} \to \frac{\tau_0}{2}$ uniformly, and the Borel transform $\mathcal{B}_G(\zeta)$ of h_G is holomorphic on $\mathbb{C}\setminus[-\frac{\tau_0}{2},\frac{\tau_0}{2}]$ with $\mathcal{B}_G(-\zeta) = -\mathcal{B}_G(\zeta)$. Moreover, for every $\varepsilon > 0$ and counterclockwise stadium contour $\Gamma_{\tau_0,\varepsilon}$ at distance ε around $[-\frac{\tau_0}{2},\frac{\tau_0}{2}]$,*

$$F_G(z) := G(-\partial_z^2)\Xi_{\mathrm{DH}}(z) = \frac{1}{2\pi\mathrm{i}}\oint_{\Gamma_{\tau_0,\varepsilon}} \mathcal{B}_G(\zeta)\,\xi_{\mathrm{DH}}\big(s(z)+\zeta\big)\,\mathrm{d}\zeta. \tag{2.4}$$

(ii) *Fix $\eta \in (0,\frac{1}{16})$, set $\varepsilon := \frac{\eta}{16}$, and let $\Gamma_+ := \{\zeta \in \Gamma_{\tau_0,\varepsilon} : \mathrm{Re}\,\zeta \geq \frac{\tau_0}{2} - \eta\}$. Define $\Omega(z) := \frac{1}{2}\log\frac{5(s(z)+1)}{2\pi}$ (principal branch on $\mathrm{Im}\,s(z) > 0$). Then uniformly for all $z = 2\gamma - 2\mathrm{i}\delta$ with $|\delta| \leq 2$ and $\mathrm{Re}\,s(z) + \frac{\tau_0}{2} \geq 1 + 3\eta$, as $\gamma \to +\infty$,*

$$I_+(z) := \frac{1}{2\pi\mathrm{i}}\int_{\Gamma_+} \mathcal{B}_G(\zeta)\,\xi_{\mathrm{DH}}\big(s(z)+\zeta\big)\,\mathrm{d}\zeta = \Gamma_{\mathrm{DH}}(s(z))\,h_G(\Omega(z))\,\big[\mathcal{D}_{\mathrm{DH}}\big(s(z)+\tfrac{\tau_0}{2}\big) + r_+(z)\big], \tag{2.5}$$

where $\sup\{|r_+(z)| : |\delta| \leq 2,\ \mathrm{Re}\,s(z) + \frac{\tau_0}{2} \geq 1+3\eta,\ \gamma \geq \Gamma\} \to 0$ as $\Gamma \to \infty$.

Proof. (i) For $\mathrm{Re}\,w > 0$, $h_G(w) \neq 0$ and $\frac{h_G'(w)}{h_G(w)} = \sum_{j=1}^\infty \frac{2w}{4w_j+w^2} = \frac{1}{2}S_G(\frac{w}{2})$. Because S_G is continuous on $[0,\infty)$ with $S_G(0) = 0$ and $\lim_{u\to\infty} S_G(u) = \tau_0$, we have $M_G := \sup_{u\geq 0} S_G(u) < \infty$. On the sector $|\arg\zeta| \leq \frac{3\pi}{8}$, $|4w_j + R^2\zeta^2| \geq \sin(\frac{\pi}{8})(4w_j + R^2|\zeta|^2)$ for every $R > 0$, so the holomorphic family $q_R(\zeta) := \frac{h_G'(R\zeta)}{h_G(R\zeta)}$ satisfies $|q_R(\zeta)| \leq \frac{1}{2\sin(\pi/8)}S_G(\frac{R|\zeta|}{2}) \leq \frac{M_G}{2\sin(\pi/8)}$ on $\{\frac{1}{2} \leq |\zeta| \leq 2,\ |\arg\zeta| \leq \frac{3\pi}{8}\}$. Since $\lim_{R\to\infty} q_R(x) = \frac{\tau_0}{2}$ for every real $x \in [\frac{1}{2},2]$, Vitali's normal-family theorem implies $q_R(\zeta) \to \frac{\tau_0}{2}$ uniformly on $\{|\zeta| = 1,\ |\arg\zeta| \leq \frac{\pi}{3}\}$, proving the first claim. Writing $S_G(u) = \int_0^\infty \frac{2u}{t+u^2}\,\mathrm{d}n_G(t)$ with $n_G(t) := \#\{j : w_j \leq t\}$, Karamata's Tauberian theorem yields $t^{-1/2}n_G(t) \to \frac{\tau_0}{\pi}$, so h_G has indicator $\frac{\tau_0}{2}|\cos\theta|$ [17, Ch. II, Thm. 2] and conjugate indicator diagram $[-\frac{\tau_0}{2},\frac{\tau_0}{2}]$ [4, Thm. 5.4.1]; evenness of h_G gives $\mathcal{B}_G(-\zeta) = -\mathcal{B}_G(\zeta)$. By Pólya's representation $h_G(2u) = \frac{1}{2\pi\mathrm{i}}\oint_{\Gamma_{\tau_0,\varepsilon}} \mathcal{B}_G(\zeta)\,\mathrm{e}^{2u\zeta}\,\mathrm{d}\zeta$ and $\xi_{\mathrm{DH}}(s(z)+\zeta) = \int_{-\infty}^\infty \Phi_{\mathrm{DH}}(u)\,\mathrm{e}^{2u\zeta+\mathrm{i}zu}\,\mathrm{d}u$, compactness of $\Gamma_{\tau_0,\varepsilon}$ and the double-exponential decay of $\Phi_{\mathrm{DH}}(u)$ allow Fubini's theorem to exchange the two integrals, establishing (2.4).

(ii) Set $\sigma_z := \mathrm{Re}\,s(z) = \frac{1}{2} + \delta \in [-\frac{3}{2},\frac{5}{2}]$. Since $\Omega(z) = \frac{1}{2}\log\frac{5\gamma}{2\pi} + \frac{\mathrm{i}\pi}{4} + \mathcal{O}(\gamma^{-1})$ lies inside $|\arg w| \leq \frac{\pi}{4}$ for $\gamma \geq \gamma_0$, integrating $\frac{h_G'(w)}{h_G(w)} = \frac{\tau_0}{2} + o(1)$ gives $|h_G(\Omega(z))| \geq \gamma^{\frac{\tau_0}{4}-\frac{\eta}{16}}$ for $\gamma \geq \gamma_1(G,\eta)$. On $\zeta \in \Gamma_+$, we have $\mathrm{Re}(s(z)+\zeta) \geq \sigma_z + \frac{\tau_0}{2} - \eta \geq 1 + 2\eta$. Set $N_\gamma := \lfloor\gamma^{1/4}\rfloor$ and split $\mathcal{D}_{\mathrm{DH}}(s(z)+\zeta) = \sum_{n=1}^{N_\gamma} a_{\mathrm{DH}}(n)n^{-s(z)-\zeta} + R_{\mathrm{Dir}}(z,\zeta)$. Because $|R_{\mathrm{Dir}}(z,\zeta)| \leq \sum_{n>N_\gamma} n^{-\sigma_z-\mathrm{Re}\,\zeta} \ll_\eta \gamma^{\frac{1}{4}(1-\sigma_z-\mathrm{Re}\,\zeta)}$ and $|\Gamma_{\mathrm{DH}}(s(z)+\zeta)/\Gamma_{\mathrm{DH}}(s(z))| \ll_G \gamma^{\frac{1}{2}\mathrm{Re}\,\zeta}$, on $\zeta \in \Gamma_+$ ($\mathrm{Re}\,\zeta \leq \frac{\tau_0}{2}+\varepsilon$) the product exponent is $\frac{1}{4}\mathrm{Re}\,\zeta + \frac{1}{4}(1-\sigma_z) \leq \frac{\tau_0}{4} + \frac{1}{4}(1+\frac{\eta}{16} - \sigma_z - \frac{\tau_0}{2}) < \frac{\tau_0}{4} - \frac{5}{8}\eta$, so the R_{Dir} integral contributes $\mathcal{O}_{G,\eta}(\gamma^{-\eta/2})|\Gamma_{\mathrm{DH}}(s(z))\,h_G(\Omega(z))|$.

For $1 \le n \le N_\gamma$, Stirling's expansion at $\frac{s(z)+1}{2}$ gives $\Gamma_{\mathrm{DH}}(s(z)+\zeta)/\Gamma_{\mathrm{DH}}(s(z)) = \mathrm{e}^{\Omega(z)\zeta}(1 + R_{\mathrm{St}}(z,\zeta))$ with $|R_{\mathrm{St}}(z,\zeta)| \ll_G \gamma^{-1}$ uniformly on $\Gamma_{\tau_0,\varepsilon}$. Summing $|n^{-s(z)-\zeta}\mathrm{e}^{\Omega(z)\zeta}R_{\mathrm{St}}| \ll_G \gamma^{-1+\frac{\tau_0}{4}+\frac{\varepsilon}{2}} n^{-(1+2\eta)}$ over $1 \le n \le N_\gamma$ contributes $\mathcal{O}_{G,\eta}(\gamma^{-1/2})|h_G(\Omega(z))|$. Next, for $1 \le n \le N_\gamma \le \gamma^{1/4}$ and $\zeta \in \Gamma_{\tau_0,\varepsilon} \setminus \Gamma_+$ ($\operatorname{Re}\zeta - \frac{\tau_0}{2} \le -\eta$, $|\operatorname{Im}\zeta| \le \varepsilon$), we have $\operatorname{Re}(\Omega(z) - \log n) \ge \frac{1}{4}\log\gamma - C_1 > 0$, whence

$$\left|n^{-s(z)}\mathrm{e}^{(\Omega(z)-\log n)\zeta}\right| \ll n^{-(\sigma_z+\tau_0/2)}\,\mathrm{e}^{(\operatorname{Re}\Omega(z)-\log n)\frac{\tau_0}{2}}\,\mathrm{e}^{-\eta(\frac{1}{4}\log\gamma - C_1)} \ll_\eta n^{-(1+3\eta)}\,\gamma^{\frac{\tau_0}{4}-\frac{\eta}{4}}.$$

Re-attaching $\Gamma_{\tau_0,\varepsilon}\setminus\Gamma_+$ therefore incurs an error $\mathcal{O}_{G,\eta}(\gamma^{-3\eta/16})|h_G(\Omega(z))|$ and evaluates each contour integral to $\frac{1}{2\pi\mathrm{i}}\oint_{\Gamma_{\tau_0,\varepsilon}} \mathcal{B}_G(\zeta)\,\mathrm{e}^{(\Omega(z)-\log n)\zeta}\,\mathrm{d}\zeta = h_G(\Omega(z)-\log n)$, yielding

$$\frac{I_+(z)}{\Gamma_{\mathrm{DH}}(s(z))\,h_G(\Omega(z))} = \sum_{n=1}^{N_\gamma} a_{\mathrm{DH}}(n)\,n^{-s(z)}\,\frac{h_G(\Omega(z)-\log n)}{h_G(\Omega(z))} + \mathcal{O}_{G,\eta}(\gamma^{-3\eta/16}). \tag{2.6}$$

(Note that for $G_t(w) = \mathrm{e}^{tw}$, $h_{G_t}(w) = \mathrm{e}^{tw^2/4}$ gives $h_{G_t}(\Omega - \log n)/h_{G_t}(\Omega) = n^{-\frac{t}{2}\Omega}\mathrm{e}^{\frac{t}{4}\log^2 n}$, recovering Dobner's modified Dirichlet series [10].) Finally, let $\delta_G(\gamma) := \sup\{|\frac{h_G'(w)}{h_G(w)} - \frac{\tau_0}{2}| : \operatorname{Re} w \ge \frac{1}{4}\log\gamma - C_1,\ |\arg w| \le \frac{\pi}{3}\}$; by (i), $\lim_{\gamma\to\infty}\delta_G(\gamma) = 0$. For $1 \le n \le N_\gamma$ and γ large enough that $\delta_G(\gamma) \le \eta$, integrating along $[\Omega(z) - \log n, \Omega(z)]$ gives $h_G(\Omega(z) - \log n)/h_G(\Omega(z)) = n^{-\tau_0/2+\theta_n(z)}$ with $|\theta_n(z)| \le \delta_G(\gamma)$. Given any $\varepsilon_1 > 0$, choosing M_1 so $\sum_{n>M_1} n^{-(1+2\eta)} < \varepsilon_1/3$ bounds both tails $n > M_1$ uniformly by $\varepsilon_1/3$, while on $1 \le n \le M_1$ we have $\sum_{n=1}^{M_1}|n^{\theta_n(z)} - 1| < \varepsilon_1/3$ once $\delta_G(\gamma)$ is sufficiently small, proving (2.5). □

Theorem 2.6 (Sufficient Stieltjes shift conditions on $\mathcal{G}_0$)**.** *Under the hypotheses of Theorem 1.1, assertions* (i)(a), (i)(b)*, and the positive half of* (i)(c) *hold for every $G \in \mathcal{G}_0$.*

Proof. In (a), $\frac{2u}{w_j+u^2} \le w_j^{-1/2}$ for all $u > 0$, and $\sum_{j=1}^\infty w_j^{-1/2} < \infty$ gives $\lim_{u\to\infty} S_G(u) = 0$ by dominated convergence. In (c), if $\lambda > 0$ then $S_G(u) \ge 4\lambda u \to +\infty$; if $\lambda = 0$ and $\underline{\rho}(G) > \frac{1}{2}$, fix $\alpha \in (\frac{1}{2}, \underline{\rho}(G))$. Since $u \mapsto u\,S_G(u) = \sum_{j=1}^\infty \frac{2u^2}{w_j+u^2}$ is strictly increasing on $(0,\infty)$,

$$\log G(u^2) - \log G(1) = \int_1^u \frac{y\,S_G(y)}{y}\,\mathrm{d}y \le u\,S_G(u)\log u \implies S_G(u) \ge \frac{u^{2\alpha} - \mathcal{O}(1)}{u\log u} \longrightarrow +\infty \tag{2.7}$$

as $u \to +\infty$, because $2\alpha > 1$.

Assume now that $\lambda = 0$ and $\lim_{u\to\infty} S_G(u) = \tau_0 \in [0,\infty)$, covering both (a) ($\tau_0 = 0$) and (b) ($\tau_0 = \tau > 0$).

Case 1: Subcritical persistence ($0 \le \tau_0 < 2\sigma^* - 1$). Set $\sigma_{\mathrm{low}} := \max(1, \frac{1}{2} + \frac{\tau_0}{2}) < \sigma^*$. Applying Lemma 2.4(ii) provides a zero $\rho_* = \beta_0 + \mathrm{i}\gamma_*$ with $\beta_0 \in (\sigma_{\mathrm{low}}, \sigma^*]$, $\eta_0 = \frac{1}{4}(\beta_0 - \sigma_{\mathrm{low}}) \in (0, \frac{1}{16})$ (shrinking η_0 if necessary), $r_0 \in (0, \min(\eta_0/2, 1/4))$, $\mu_0 > 0$, and $M(T) \gg_G T$ unit-separated shifts $\tau_m \in [\frac{1}{2}T, \frac{3}{4}T]$ (T large) satisfying (2.3) on $|w - s_m| = r_0$, where $s_m := \rho_* + \mathrm{i}\tau_m = \beta_0 + \mathrm{i}\gamma_m$ ($\gamma_m := \gamma_* + \tau_m \in [\frac{1}{3}T, T]$). Set $z_m^{\mathrm{sh}} := 2\gamma_m - 2\mathrm{i}(\beta_0 - \frac{1}{2} - \frac{\tau_0}{2})$, so $s(z_m^{\mathrm{sh}}) + \frac{\tau_0}{2} = s_m$ and $|s(z) + \frac{\tau_0}{2} - s_m| = \frac{1}{2}|z - z_m^{\mathrm{sh}}|$. The disks $|z - z_m^{\mathrm{sh}}| \le 2r_0$ are pairwise disjoint and satisfy $\operatorname{Im} z \le -2(\beta_0 - \frac{1}{2} - \frac{\tau_0}{2}) + 2r_0 \le -7\eta_0 < 0$ (and $\operatorname{Im} z < -1$ when $\tau_0 = 0$, since $\beta_0 - r_0 > 1$). Moreover, on $|z - z_m^{\mathrm{sh}}| \le 2r_0$ we have $\sigma_z := \operatorname{Re} s(z) \ge \beta_0 - \frac{\tau_0}{2} - r_0 > \frac{1}{2} + \frac{7}{2}\eta_0$ and $\sigma_z + \frac{\tau_0}{2} > 1 + \frac{7}{2}\eta_0$. Applying Lemma 2.5 with $\eta = \eta_0$ and $\varepsilon = \frac{\eta_0}{16}$ gives (2.5) on Γ_+. On $\zeta \in \Gamma_{\tau_0,\varepsilon}\setminus\Gamma_+$ ($-\frac{\tau_0}{2} - \varepsilon \le \operatorname{Re}\zeta \le \frac{\tau_0}{2} - \eta_0$), Lemma 2.4(i) with parameter $\frac{\eta_0}{2}$ bounds $|\xi_{\mathrm{DH}}(s(z)+\zeta)| \ll_{G,\eta_0} \gamma_m^{\frac{1}{2}\max(\sigma, 1-\sigma, 1+\eta_0/2)}\mathrm{e}^{-\frac{\pi}{4}\gamma_m}$, where $\sigma = \sigma_z + \operatorname{Re}\zeta$ satisfies

$$\left(\frac{1}{2}\sigma_z + \frac{\tau_0}{4}\right) - \frac{1}{2}\max\Big(\sigma,\, 1-\sigma,\, 1+\frac{\eta_0}{2}\Big) = \min\Big(\tfrac{1}{2}(\tfrac{\tau_0}{2} - \operatorname{Re}\zeta),\ \sigma_z - \tfrac{1}{2} + \tfrac{1}{2}(\tfrac{\tau_0}{2} + \operatorname{Re}\zeta),$$
$$\tfrac{1}{2}(\sigma_z + \tfrac{\tau_0}{2} - 1 - \tfrac{\eta_0}{2})\Big) \ge \frac{\eta_0}{2},$$

so $\Gamma_{\tau_0,\varepsilon}\setminus\Gamma_+$ contributes $\mathcal{O}_{G,\eta_0}(\gamma_m^{-7\eta_0/16})|\Gamma_{\mathrm{DH}}(s(z))\,h_G(\Omega(z))|$. Hence uniformly on $|z-z_m^{\mathrm{sh}}|=2r_0$, $F_G(z)/[\Gamma_{\mathrm{DH}}(s(z))\,h_G(\Omega(z))]=\mathcal{D}_{\mathrm{DH}}(s(z)+\frac{\tau_0}{2})+o_G(1)$, whereas $|\mathcal{D}_{\mathrm{DH}}(s(z)+\frac{\tau_0}{2})|>\frac{\mu_0}{2}$ by (2.3). For T sufficiently large, Rouché's theorem yields at least one zero of F_G inside each disk $|z-z_m^{\mathrm{sh}}|<2r_0\subset\{\operatorname{Im}z<0\}$, proving $N_{\mathrm{off}}(F_G;T)\gg_G T$.

Case 2: Supercritical finiteness ($\tau_0=\tau>2\sigma^*-1$). By $F_G(-z)=F_G(z)$, $F_G(\bar z)=\overline{F_G(z)}$, and Lemma 2.1(i), every zero satisfies $|\delta|\le h_\infty^*$ ($z=x-2\mathrm{i}\delta$, $x=2\gamma$), and it suffices to find $X_0>0$ such that all zeros of $F_G(z)$ in $\operatorname{Re}z\ge X_0$ are real. Choose $\eta_2\in(0,\min(\frac{1}{16},\frac{1}{6}(\frac12+\frac{\tau}{2}-\sigma^*)))$ and $\varepsilon:=\frac{\eta_2}{16}$. For all $\delta\in[-2\eta_2,h_\infty^*]$ and $\gamma\ge\gamma_1$, we have $\operatorname{Re}(s(z)+\frac{\tau}{2})=\frac12+\delta+\frac{\tau}{2}\ge\sigma^*+4\eta_2>1+4\eta_2$, so Lemma 2.4(iii) gives $|\mathcal{D}_{\mathrm{DH}}(s(z)+\frac{\tau}{2})|\ge c(4\eta_2)>0$. Consequently,

$$\mathcal{M}_G(z):=\Gamma_{\mathrm{DH}}(s(z))\,h_G(\Omega(z))\,\mathcal{D}_{\mathrm{DH}}(s(z)+\tfrac{\tau}{2}) \tag{2.8}$$

is holomorphic and non-vanishing on $\{\gamma\ge\gamma_1,\ \delta\in[-2\eta_2,h_\infty^*]\}$, and Lemma 2.5(ii) gives $I_+(z)=\mathcal{M}_G(z)(1+o(1))$ uniformly there as $\gamma\to+\infty$.

First, on the outer strip $\delta\in[\frac{\eta_2}{4},h_\infty^*]$ ($\sigma_z-\frac12=\delta\ge\frac{\eta_2}{4}$), Lemma 2.4(i) with parameter $\frac{\eta_2}{2}$ bounds $\Gamma_{\tau,\varepsilon}\setminus\Gamma_+$ by $\mathcal{O}_{G,\eta_2}(\gamma^{-5\eta_2/32})|\mathcal{M}_G(z)|$ (since $\delta+\frac12(\frac{\tau}{2}+\operatorname{Re}\zeta)\ge\frac{\eta_2}{4}-\frac{\eta_2}{32}=\frac{7}{32}\eta_2$ and $|h_G(\Omega(z))|\ge\gamma^{\frac{\tau}{4}-\frac{\eta_2}{16}}$), so $F_G(z)=\mathcal{M}_G(z)(1+o(1))\ne0$ for $\gamma\ge\gamma_2$, and by conjugation $F_G(z)\ne0$ on $[-h_\infty^*,-\frac{\eta_2}{4}]$.

Second, on the remaining strip $|\delta|\le\frac{\eta_2}{4}$, split $\Gamma_{\tau,\varepsilon}=\Gamma_+\cup\Gamma_-\cup\Gamma_{\mathrm{mid}}$, where $\Gamma_-:=-\Gamma_+$ ($\operatorname{Re}\zeta\le-\frac{\tau}{2}+\eta_2$) and $\Gamma_{\mathrm{mid}}:=\{\zeta\in\Gamma_{\tau,\varepsilon}:|\operatorname{Re}\zeta|<\frac{\tau}{2}-\eta_2\}$. Because $\mathcal{B}_G(-\zeta)=-\mathcal{B}_G(\zeta)$, $\mathcal{B}_G(\bar w)=\overline{\mathcal{B}_G(w)}$, and $\xi_{\mathrm{DH}}(s(z)-u)=\xi_{\mathrm{DH}}(1-s(z)+u)=\overline{\xi_{\mathrm{DH}}(s(\bar z)+\bar u)}$, substituting $\zeta=-\bar w$ yields the exact identity $I_-(z)=\overline{I_+(\bar z)}$. Since both z and $\bar z$ have ordinate in $[-\frac{\eta_2}{4},\frac{\eta_2}{4}]\subset[-2\eta_2,h_\infty^*]$, we obtain $I_+(z)+I_-(z)=\mathcal{M}_G(z)(1+o(1))+\overline{\mathcal{M}_G(\bar z)}(1+o(1))$, while Lemma 2.4(i) on Γ_{mid} gives $|I_{\mathrm{mid}}(z)|\ll_{G,\eta_2}\gamma^{-7\eta_2/16}\max(|\mathcal{M}_G(z)|,|\overline{\mathcal{M}_G(\bar z)}|)$. Thus uniformly on $|\delta|\le\frac{\eta_2}{4}$ as $\gamma\to+\infty$,

$$F_G(z)=H_{\mathrm{main}}(z)+E_G(z),\qquad H_{\mathrm{main}}(z):=\mathcal{M}_G(z)+\overline{\mathcal{M}_G(\bar z)}, \tag{2.9}$$

where $|E_G(z)|=o_{G,\eta_2}(1)\cdot\max(|\mathcal{M}_G(z)|,|\overline{\mathcal{M}_G(\bar z)}|)$. Differentiating $\log\mathcal{M}_G(z)$ on $|\delta|\le\frac{\eta_2}{2}$ gives: $\frac{\mathrm{d}}{\mathrm{d}z}\log\Gamma_{\mathrm{DH}}(s(z))=\frac{\mathrm{i}}{4}[\log\frac{5}{\pi}+\psi(\frac{s(z)+1}{2})]=-\frac{\pi}{8}+\frac{\mathrm{i}}{4}\log\frac{5\gamma}{2\pi}+\mathcal{O}(\gamma^{-1})$; $\frac{\mathrm{d}}{\mathrm{d}z}\log h_G(\Omega(z))=\frac{\mathrm{i}}{4(s(z)+1)}(\frac{\tau}{2}+o(1))=\mathcal{O}_G(\gamma^{-1})$; and $|\frac{\mathrm{d}}{\mathrm{d}z}\log\mathcal{D}_{\mathrm{DH}}(s(z)+\frac{\tau}{2})|\le\frac{-\zeta'(1+4\eta_2)}{2c(4\eta_2)}=\mathcal{O}_{\eta_2}(1)$. Hence uniformly on $|\delta|\le\frac{\eta_2}{2}$, $\gamma\ge\gamma_2$,

$$\frac{\mathrm{d}}{\mathrm{d}z}\log\mathcal{M}_G(z)=-\frac{\pi}{8}+\frac{\mathrm{i}}{4}\log\frac{5\gamma}{2\pi}+\mathcal{O}_{G,\eta_2}(1). \tag{2.10}$$

For fixed $x=2\gamma$, $\frac{\partial}{\partial u}\log|\mathcal{M}_G(x-2\mathrm{i}u)|=2\operatorname{Im}(\frac{\mathcal{M}_G'}{\mathcal{M}_G}(x-2\mathrm{i}u))=\frac12\log\frac{5\gamma}{2\pi}+\mathcal{O}_{G,\eta_2}(1)\ge\frac38\log\gamma$ for $\gamma\ge\gamma_3$, so integrating over $[-\delta,\delta]$ yields $\log|\mathcal{M}_G(x-2\mathrm{i}\delta)/\overline{\mathcal{M}_G(x+2\mathrm{i}\delta)}|\ge\frac34\delta\log\gamma>0$ for every $\delta\in(0,\frac{\eta_2}{4}]$. This has two consequences for $x=2\gamma\ge X_0$ (X_0 chosen so $|E_G(z)|<\frac14\max(|\mathcal{M}_G(z)|,|\overline{\mathcal{M}_G(\bar z)}|)$ on $|\delta|\le\frac{\eta_2}{4}$):

(a) $H_{\mathrm{main}}(z)$ has no non-real zeros in $\{x\ge X_0,\ |\delta|\le\frac{\eta_2}{4}\}$, and whenever $\frac{4}{\log(x/2)}\le|\delta|\le\frac{\eta_2}{4}$, we have $|H_{\mathrm{main}}(z)|\ge(1-\mathrm{e}^{-3})\max(|\mathcal{M}_G(z)|,|\overline{\mathcal{M}_G(\bar z)}|)>\frac{9}{10}\max(|\mathcal{M}_G(z)|,|\overline{\mathcal{M}_G(\bar z)}|)>|E_G(z)|$, excluding zeros of $F_G(z)$ in $\frac{4}{\log(x/2)}\le|\delta|\le\frac{\eta_2}{4}$.

(b) On $\mathbb{R}\cap[X_0,\infty)$, writing $\mathcal{M}_G(x)=|\mathcal{M}_G(x)|\mathrm{e}^{\mathrm{i}\vartheta(x)}$ gives $\vartheta'(x)=\operatorname{Im}(\frac{\mathcal{M}_G'(x)}{\mathcal{M}_G(x)})=\frac14\log\frac{5x}{4\pi}+\mathcal{O}_{G,\eta_2}(1)>\frac18\log x>0$, so the points x_k ($k\ge k_0$) with $\vartheta(x_k)=k\pi$ satisfy $x_{k+1}-x_k\asymp\frac{4\pi}{\log x_k}$ and separate the simple real zeros $r_k\in(x_k,x_{k+1})$ of $H_{\mathrm{main}}(x)=2|\mathcal{M}_G(x)|\cos\vartheta(x)$. Set $\delta_k:=\frac{4}{\log(x_k/2)}$ and $\mathcal{R}_k:=[x_k,x_{k+1}]\times[-2\mathrm{i}\delta_k,2\mathrm{i}\delta_k]$. Because $\delta_k\ge\frac{4}{\log(x/2)}$ for all $x\in[x_k,x_{k+1}]$, the union $\bigcup_{k\ge k_0}\mathcal{R}_k$ covers $\{x\ge x_{k_0},\ |\delta|\le\frac{4}{\log(x/2)}\}$ with no corner gap, and (a) already gives $|H_{\mathrm{main}}(z)|>|E_G(z)|$ along both horizontal edges $z=x\mp2\mathrm{i}\delta_k$. Along the vertical edges $z=x_j-2\mathrm{i}\delta$ ($j\in\{k,k+1\}$, $|\delta|\le\delta_k$), integrating (2.10) from x_j gives $\mathcal{M}_G(x_j-2\mathrm{i}\delta)=(-1)^j|\mathcal{M}_G(x_j-2\mathrm{i}\delta)|\mathrm{e}^{\mathrm{i}\frac{\pi}{4}\delta+\mathcal{O}(1/\log x_k)}$ and $\overline{\mathcal{M}_G(x_j+2\mathrm{i}\delta)}=(-1)^j|\mathcal{M}_G(x_j+$

$2\mathrm{i}\delta)|\mathrm{e}^{\mathrm{i}\frac{\pi}{4}\delta+\mathcal{O}(1/\log x_k)}$; since the two summands form an angle $\mathcal{O}(1/\log x_k) < \frac{\pi}{6}$, we have $|H_{\mathrm{main}}(z)| \geq \max(|\mathcal{M}_G(z)|, |\overline{\mathcal{M}_G(\bar{z})}|) > |E_G(z)|$ on both vertical edges as well.

By Rouché's theorem, F_G has a unique zero (counted with multiplicity) inside $\mathrm{int}(\mathcal{R}_k)$ for every $k \geq k_0$; by conjugation symmetry $F_G(\bar{z}) = \overline{F_G(z)}$ of $\mathcal{R}_k$, this unique zero must be real and simple (confirming also via Cauchy's estimate $|E_G'(x)| = o(\log x)|\mathcal{M}_G(x)|$ on disks of radius $\frac{4}{\log(x/2)}$ that F_G' does not vanish there). Thus $N_{\mathrm{off}}(F_G;\infty) < \infty$.

Finally, in (c), if $\lambda > 0$, write $G(w) = \mathrm{e}^{\lambda w}G_1(w)$ with $G_1 \in \mathcal{G}_0$; Proposition 4.3 gives $N_{\mathrm{off}}(H_\lambda^{\mathrm{DH}};\infty) < \infty$, and Lemma 2.2 yields $N_{\mathrm{off}}(G(-\partial_z^2)\Xi_{\mathrm{DH}};\infty) \leq N_{\mathrm{off}}(H_\lambda^{\mathrm{DH}};\infty) < \infty$. If $\lambda = 0$ and $w_{j+1} - w_j = o(\sqrt{w_j})$, set $\tau_1 := 2\sigma^* > 2\sigma^* - 1$ and $t_m := (\frac{\pi m}{\tau_1})^2$. Since $t_{m+1} - t_m = \frac{2\pi}{\tau_1}\sqrt{t_m} + \frac{\pi^2}{\tau_1^2}$ dominates $o(\sqrt{t_m})$, every open interval (t_m, t_{m+1}) $(m \geq m_0)$ contains at least one root w_{j_m}. Disjointness of (t_m, t_{m+1}) ensures $j_{m_0} < j_{m_0+1} < \cdots$ and $w_{j_m}/t_m \to 1$, so $G_{\mathrm{sub}}(w) := \prod_{m=m_0}^{\infty}(1 + w/w_{j_m})$ satisfies $\lim_{u\to\infty} S_{G_{\mathrm{sub}}}(u) = \tau_1 > 2\sigma^* - 1$, giving $N_{\mathrm{off}}(G_{\mathrm{sub}}(-\partial_z^2)\Xi_{\mathrm{DH}};\infty) < \infty$, and Lemma 2.2 applied to $G/G_{\mathrm{sub}} \in \mathcal{G}_0$ completes the proof. □

Proposition 2.7 (Lacunary canonical product counterexample)**.** *Let $\rho \in (1/2, 1)$, define recursively $R_1 := 16$ and $R_k := 2^{2^{R_{k-1}}}$ for $k \geq 2$, and set $M_k := \lfloor R_k^{2\rho} \rfloor$. Then $G_{\mathrm{lac}}(w) := \prod_{k=1}^{\infty}(1 + w/R_k^2)^{M_k} \in \mathcal{G}_0$ has upper order $\rho(G_{\mathrm{lac}}) = \rho$ and lower order $\underline{\rho}(G_{\mathrm{lac}}) = 0$, satisfies $\lim_{k\to\infty} S_{G_{\mathrm{lac}}}(u_k) = 0$ along $u_k := R_k^{(1-\rho)/2}$, and $G_{\mathrm{lac}}(-\partial_z^2)\Xi_{\mathrm{DH}}(z)$ possesses infinitely many non-real zeros inside $|\operatorname{Im} z| > 1$.*

Proof. Because $\sum_{k=1}^{\infty} M_k R_k^{-2} \leq \sum_{k=1}^{\infty} R_k^{2\rho-2} =: C_{\mathrm{lac}} < \infty$ and $n_{G_{\mathrm{lac}}}(t) \leq \sum_{R_j \leq \sqrt{t}} R_j^{2\rho} \ll t^\rho$ with $n_{G_{\mathrm{lac}}}(R_k^2) \geq M_k \sim (R_k^2)^\rho$, Borel's theorem [4, Thm. 2.6.5] gives $G_{\mathrm{lac}} \in \mathcal{G}_0$ and $\rho(G_{\mathrm{lac}}) = \rho$. Since $R_{k-1} = \log_2 \log_2 R_k$ $(k \geq 2)$, the prefix degree satisfies $D_{k-1} := \sum_{j=1}^{k-1} M_j \ll (\log\log R_k)^{2\rho}$. At $u_k = R_k^{(1-\rho)/2}$, $\log G_{\mathrm{lac}}(u_k^2) \leq D_{k-1}\log(1 + u_k^2) + \sum_{j=k}^{\infty} M_j u_k^2 R_j^{-2} \ll (\log\log R_k)^{2\rho}\log R_k + R_k^{\rho-1}$, whence $\underline{\rho}(G_{\mathrm{lac}}) = 0$, and $S_{G_{\mathrm{lac}}}(u_k) \leq \frac{2D_{k-1}}{u_k} + \sum_{j=k}^{\infty} \frac{2M_j u_k}{R_j^2} \ll (\log\log R_k)^{2\rho} R_k^{-(1-\rho)/2} + R_k^{-\frac{3}{2}(1-\rho)} \to 0$.

To produce non-real zeros in $|\operatorname{Im} z| > 1$, apply Lemma 2.4(ii) with $\sigma_{\mathrm{low}} = 1$ to obtain a fixed zero $\rho_* = \beta_0 + \mathrm{i}\gamma_*$ $(\beta_0 \in (1, \sigma^*])$, $\eta_0 = \frac{1}{4}(\beta_0 - 1)$, $r_0 \in (0, \min(\eta_0/2, 1/4))$, and $\mu_0 > 0$. For each large k, set $T_k := R_k^{(1-\rho)/4}$ and choose via Lemma 2.4(ii) a shift $\tau^{(k)} \in [T_k, 2T_k]$ so that $|\mathcal{D}_{\mathrm{DH}}(w)| > \frac{\mu_0}{2}$ on $|w - s_k| = r_0$ and $\mathcal{D}_{\mathrm{DH}}$ vanishes inside $|w - s_k| < r_0$, where $s_k := \rho_* + \mathrm{i}\tau^{(k)} = \beta_0 + \mathrm{i}\gamma_k$ $(\gamma_k \asymp T_k)$. Set $z_k := 2\gamma_k - 2\mathrm{i}(\beta_0 - \frac{1}{2})$, so $|z - z_k| \leq 2r_0$ lies inside $\operatorname{Im} z \leq -1 - 7\eta_0 < -1$, and factor $G_{\mathrm{lac}} = G_{\leq k-1}G_{\geq k}$. Shifting $u = v + \mathrm{i}\alpha_k$ with $\alpha_k := \frac{\pi}{8} - \frac{1}{\gamma_k}$ in $\Delta_k(z) := \int_{-\infty}^{\infty} G_{\leq k-1}(u^2)(G_{\geq k}(u^2) - 1)\Phi_{\mathrm{DH}}(u)\mathrm{e}^{\mathrm{i}zu}\,\mathrm{d}u$ (legitimate on $|\operatorname{Im} u| < \frac{\pi}{8}$ by the double-exponential decay of Φ_{DH}) extracts $|\mathrm{e}^{\mathrm{i}z(v+\mathrm{i}\alpha_k)}| \asymp \mathrm{e}^{-\frac{\pi}{4}\gamma_k}\mathrm{e}^{2\delta v}$. Using $|\Phi_{\mathrm{DH}}(v + \mathrm{i}\alpha_k)| \ll \gamma_k \mathrm{e}^{-|v|}$ on $|v| \leq V_k := \log\gamma_k$ and $\ll \mathrm{e}^{3|v| - \frac{2}{5}\gamma_k^{-1}\mathrm{e}^{4|v|}}$ on $|v| > V_k$, together with $|G_{\leq k-1}(u^2)| \leq (2 + V_k^2)^{D_{k-1}} = R_k^{o(1)}$ and $|G_{\geq k}(u^2) - 1| \ll (\log R_k)^2 R_k^{2\rho-2}$ on $|v| \leq V_k$, dividing by $|\Gamma_{\mathrm{DH}}(s(z))| \geq \gamma_k^{1/2}\mathrm{e}^{-\frac{\pi}{4}\gamma_k}$ yields $|\Delta_k(z)/\Gamma_{\mathrm{DH}}(s(z))| \ll R_k^{2\rho-2+\frac{1-\rho}{4}(2\sigma^*-1/2)+o(1)} \leq R_k^{-\frac{3}{2}(1-\rho)+o(1)} \to 0$ (as $2\sigma^* - \frac{1}{2} < 2$). Meanwhile, for the polynomial $G_{\leq k-1}$ of degree $D_{k-1} \ll (\log\log\gamma_k)^{2\rho}$, writing $\xi_{\mathrm{DH}}(s + \zeta)/\Gamma_{\mathrm{DH}}(s) = \mathrm{e}^{\Omega(z)\zeta} f_s(\zeta)$ and applying Cauchy's bounds $|f_s^{(j)}(0)|/\Gamma(j + 1) \leq C_2(\eta_0/4)^{-j}$ on $|\zeta| \leq \eta_0/4$ alongside $G_{\leq k-1}(\frac{1}{4}|\Omega(z)|^2) \leq \mathrm{e}^{\frac{1}{2}C_{\mathrm{lac}}}|G_{\leq k-1}(\frac{1}{4}\Omega(z)^2)|$ and $|G_{\leq k-1}(\frac{1}{4}\Omega(z)^2)| \geq 1$ gives $G_{\leq k-1}(-\partial_z^2)\Xi_{\mathrm{DH}}(z)/[\Gamma_{\mathrm{DH}}(s(z))G_{\leq k-1}(\frac{1}{4}\Omega(z)^2)] = \mathcal{D}_{\mathrm{DH}}(s(z)) + \mathcal{O}(D_{k-1}/\log\gamma_k)$ on $|z - z_k| = 2r_0$. Rouché's theorem produces a zero of $G_{\mathrm{lac}}(-\partial_z^2)\Xi_{\mathrm{DH}}$ in $|z - z_k| < 2r_0 \subset \{\operatorname{Im} z < -1\}$ for every large k. □

Remark 2.8 (Open boundary regimes). Theorem 2.6 and Proposition 2.7 provide sufficient conditions rather than an exhaustive characterization of $\mathcal{G}_0$. Three boundary regimes remain open:

(a) multipliers with $\lambda = 0$ whose Stieltjes derivative oscillates across the threshold, $\liminf_{u\to\infty} S_G(u) < 2\sigma^* - 1 < \limsup_{u\to\infty} S_G(u)$, without root-gap regularity;
(b) the critical boundary case $\lambda = 0$ and $\lim_{u\to\infty} S_G(u) = 2\sigma^* - 1$, where the shifted abscissa $\frac{1}{2} + \frac{\tau_0}{2} = \sigma^*$ meets the supremum of off-line zeros without a uniform positive lower bound on $|\mathcal{D}_{\mathrm{DH}}|$ (for $G_{\mathrm{sinc},\sigma^*-1/2}$, Proposition 3.3 proves $8 \le N_{\mathrm{off}} < \infty$ conditionally on RH for $\zeta(s)$ and GRH for $L(s,\chi_{5,\mathrm{quad}})$, while Remark 3.4 records the unconditional Vinogradov–Korobov and Baker–Wüstholz barriers); and
(c) multipliers with $\lambda = 0$ and lower order $\underline{\rho}(G) > \frac{1}{2}$ (so $\lim_{u\to\infty} S_G(u) = +\infty$) whose roots contain clusters violating $w_{j+1} - w_j = o(\sqrt{w_j})$ and admitting no supercritical regular subproduct.

3. Global versus Asymptotic Threshold Separation for Pólya's Sinc Multiplier

Both $G_{\cosh,h}(w) = \cosh(2h\sqrt{w})$ and Pólya's hyperbolic sinc multiplier $G_{\mathrm{sinc},h}(w) = \frac{\sinh(2h\sqrt{w})}{2h\sqrt{w}} = \prod_{j=1}^{\infty}(1 + \frac{4h^2 w}{\pi^2 j^2})$ belong to $\mathcal{G}_0$ with identical asymptotic shift $\lim_{u\to\infty} S_G(u) = 2h$ and identical infinite-height threshold $h^*_\infty = \sigma^* - \frac{1}{2}$. In this section we prove Theorem 1.1(ii) in a manner completely independent of Theorem 2.6.

Lemma 3.1 (Enclosure of σ^*, strict zero inequality, and global threshold of $G_{\cosh,h}$)**.** *The unique root $\sigma^* > 1$ of the Bombieri–Ghosh prime-angle identity $\sum_{p\equiv 2,3 \pmod 5} \arctan(p^{-\sigma^*}) = \frac{\pi}{2} - \arctan\kappa$ satisfies*

$$\sigma^* \in \big[1.12036249818332506180145892226464 \pm 10^{-30}\big]. \tag{3.1}$$

Moreover, every zero $\rho = \sigma + \mathrm{i}\gamma$ of $\mathcal{D}_{\mathrm{DH}}(s)$ satisfies $\sigma < \sigma^*$ strictly, every zero of $\Xi_{\mathrm{DH}}(z)$ satisfies $|\operatorname{Im} z| < 2(\sigma^* - \frac{1}{2})$, and

$$h^*_{\mathrm{global}}(G_{\cosh}; \Xi_{\mathrm{DH}}) = \sigma^* - \tfrac{1}{2} \in \Big(\frac{310}{500}, \frac{5}{8}\Big). \tag{3.2}$$

Proof. Summing $p < 1000$ directly and expanding $\arctan(p^{-\sigma}) = \sum_{m=0}^{14} \frac{(-1)^m}{2m+1} p^{-(2m+1)\sigma} + \mathcal{O}(10^{-95})$ for $p \ge 1000$ expresses each odd prime-power sum via Möbius inversion of $\frac{1}{2}\log[(1-5^{-u})\zeta(u)/L(u,\chi_{5,\mathrm{quad}})]$, where $\chi_{5,\mathrm{quad}}$ is the real non-principal character modulo 5. Evaluating at $1.12036249818332506180145892226464 \pm 5\times 10^{-31}$ in 320-bit `FLINT/Arb` arithmetic certifies opposite signs by the Intermediate Value Theorem, proving (3.1).

Next, let $\rho = \sigma + \mathrm{i}\gamma$ be any zero of $\mathcal{D}_{\mathrm{DH}}(s)$. If $\sigma \le 1$, then $\sigma < \sigma^*$. If $\sigma > 1$, then $L(s,\chi_5)$ and $L(s,\overline{\chi_5})$ do not vanish, and $\mathcal{D}_{\mathrm{DH}}(\sigma + \mathrm{i}\gamma) = 0$ is equivalent to

$$\frac{L(\sigma+\mathrm{i}\gamma, \overline{\chi_5})}{L(\sigma+\mathrm{i}\gamma, \chi_5)} = -\frac{1-\mathrm{i}\kappa}{1+\mathrm{i}\kappa} = \mathrm{e}^{+2\mathrm{i}(\frac{\pi}{2} - \arctan\kappa)}. \tag{3.3}$$

In the Euler product for (3.3), primes $p \equiv 1,4 \pmod 5$ have $\overline{\chi_5(p)} = \chi_5(p)$ and cancel identically, while each $p \equiv 2,3 \pmod 5$ contributes $\frac{1-w_p}{1+w_p}$ with $w_p := \chi_5(p) p^{-\sigma-\mathrm{i}\gamma}$ ($|w_p| = p^{-\sigma} < 1$). The principal argument satisfies $|\arg\frac{1-w_p}{1+w_p}| \le 2\arctan(p^{-\sigma})$, with equality $\arg\frac{1-w_p}{1+w_p} = +2\arctan(p^{-\sigma})$ if and only if $w_p = -\mathrm{i}p^{-\sigma}$. Since $\sum_{p\equiv 2,3 \pmod 5} 2\arctan(p^{-\sigma}) \le 2(\frac{\pi}{2} - \arctan\kappa) < \pi$ whenever $\sigma \ge \sigma^*$, equality in (3.3) at $\sigma \ge \sigma^*$ requires $\sigma = \sigma^*$ and $w_p = -\mathrm{i}p^{-\sigma^*}$ for every $p \equiv 2,3 \pmod 5$, in particular for both $p = 2$ and $p = 3$. When $\gamma = 0$, $w_2 = +\mathrm{i}2^{-\sigma} \ne -\mathrm{i}2^{-\sigma}$ gives $\arg\frac{1-w_2}{1+w_2} = -2\arctan(2^{-\sigma}) < 0$ (opposite in sign to $\arg\frac{1-w_3}{1+w_3} = +2\arctan(3^{-\sigma}) > 0$), so equality fails at $\gamma = 0$. When $\gamma \ne 0$, simultaneous equality $w_2 = \mathrm{i}2^{-\sigma-\mathrm{i}\gamma} = -\mathrm{i}2^{-\sigma}$ and $w_3 = -\mathrm{i}3^{-\sigma-\mathrm{i}\gamma} = -\mathrm{i}3^{-\sigma}$ requires $2^{-\mathrm{i}\gamma} = -1$ and $3^{-\mathrm{i}\gamma} = +1$, so $\gamma\log 2 = (2k_1+1)\pi$ and $\gamma\log 3 = 2k_2\pi$ ($k_1 \in \mathbb{Z}$, $k_2 \in \mathbb{Z}\setminus\{0\}$), contradicting $\frac{\log 3}{\log 2} \notin \mathbb{Q}$. Thus $\sigma < \sigma^*$ strictly; by $\xi_{\mathrm{DH}}(s) = \xi_{\mathrm{DH}}(1-s)$ and $\sigma^* > 1$, every zero of $\Xi_{\mathrm{DH}}(z)$ satisfies $|\operatorname{Im} z| < 2(\sigma^* - \frac{1}{2})$.

Consequently, Lemma 2.1(ii) implies that $G_{\cosh,h}(-\partial_z^2)\Xi_{\mathrm{DH}}(z)$ has only real zeros for every $h \geq \sigma^* - \frac{1}{2}$, so $h^*_{\mathrm{global}}(G_{\cosh};\Xi_{\mathrm{DH}}) \leq \sigma^* - \frac{1}{2}$. Conversely, for any $0 < h < \sigma^* - \frac{1}{2}$, write $G_{\cosh,h}(-\partial_z^2)\Xi_{\mathrm{DH}}(z) = \frac{1}{2}[\xi_{\mathrm{DH}}(s(z)+h)+\xi_{\mathrm{DH}}(s(z)-h)]$ and apply Lemma 2.4(ii) with $\sigma_{\mathrm{low}} := \max(1, \frac{1}{2}+h) < \sigma^*$ to obtain shifted disks $|s(z)+h-(\rho_*+\mathrm{i}\tau_m)| \leq r_0$ inside $\operatorname{Im} z < 0$ on whose boundary $|\mathcal{D}_{\mathrm{DH}}(s(z)+h)| > \frac{\mu_0}{2}$. By Lemma 2.4(i), $|\xi_{\mathrm{DH}}(s(z)-h)/\Gamma_{\mathrm{DH}}(s(z)+h)| \ll_h \gamma_m^{-\eta_0} \to 0$ as $\tau_m \to +\infty$, so Rouché's theorem produces non-real zeros of $G_{\cosh,h}(-\partial_z^2)\Xi_{\mathrm{DH}}$ for every $0 < h < \sigma^* - \frac{1}{2}$, proving (3.2). □

Theorem 3.2 (Global bounds and strict threshold separation for $G_{\mathrm{sinc},h}$)**.**

(i) *For every $h \in [0, \frac{5}{8}]$, the entire function $G_{\mathrm{sinc},h}(-\partial_z^2)\Xi_{\mathrm{DH}}(z)$ possesses at least one non-real zero; at $h = \frac{5}{8} = 0.625 > \sigma^* - \frac{1}{2}$, it possesses a certified simple non-real zero in the box $B_{5/8} := \{z \in \mathbb{C} : |\operatorname{Re}(z - z_{44838})| \leq 10^{-30},\ |\operatorname{Im}(z - z_{44838})| \leq 10^{-30}\}$ centered at*

$$\begin{aligned} z_{44838} = {} & 89676.898678875343559173771577620034736 \\ & - 0.010563860753475127143116220777059191 10\,\mathrm{i}, \end{aligned} \tag{3.4}$$

originating at $h = 0$ from the zero $\rho_{44838} = 0.91613758\ldots + 44838.47521067\ldots\,\mathrm{i}$ of $\mathcal{D}_{\mathrm{DH}}(s)$ (and persisting at $h = \frac{625121}{10^6} = 0.625121$).

(ii) *Conversely, for every $h \geq 2\sigma^* - 1$, all zeros of $G_{\mathrm{sinc},h}(-\partial_z^2)\Xi_{\mathrm{DH}}(z)$ are real, and all are simple whenever $h > 2\sigma^* - 1$.*

Consequently, $\sigma^* - \frac{1}{2} = h^*_{\mathrm{global}}(G_{\cosh};\Xi_{\mathrm{DH}}) < \frac{5}{8} < h^*_{\mathrm{global}}(G_{\mathrm{sinc}};\Xi_{\mathrm{DH}}) \leq 2\sigma^* - 1$.

Proof. (i) Set $s_{\mathrm{ref}} := \frac{1}{2} + 44838.46\,\mathrm{i}$, $C_{\mathrm{ref}} := \Gamma_{\mathrm{DH}}(s_{\mathrm{ref}}) \neq 0$, and $\tilde{\xi}_{\mathrm{DH}}(s) := \xi_{\mathrm{DH}}(s)/C_{\mathrm{ref}}$. By Fubini's theorem on $G_{\mathrm{sinc},h}(u^2) = \frac{1}{2}\int_{-1}^{1} \mathrm{e}^{2huv}\,\mathrm{d}v$, expanding $\tilde{\xi}_{\mathrm{DH}}(s_{\mathrm{ref}}+w) = \sum_{k=0}^{\infty} c_k w^k$ ($c_k := \tilde{\xi}^{(k)}_{\mathrm{DH}}(s_{\mathrm{ref}})/\Gamma(k+1)$) yields for all $z \in \mathbb{C}$ and all $h \in \mathbb{R}$ (including $h = 0$, with no division by h) the singularity-free power-series identity

$$\begin{aligned} g_{\mathrm{sinc}}(z,h) := C_{\mathrm{ref}}^{-1}\, G_{\mathrm{sinc},h}(-\partial_z^2)\Xi_{\mathrm{DH}}(z) &= \frac{1}{2}\int_{-1}^{1} \tilde{\xi}_{\mathrm{DH}}(s(z)+hv)\,\mathrm{d}v \\ &= \sum_{k=0}^{\infty} c_k \sum_{m=0}^{\lfloor k/2 \rfloor} \binom{k}{2m} \frac{(s(z)-s_{\mathrm{ref}})^{k-2m}}{2m+1}\, h^{2m}. \end{aligned} \tag{3.5}$$

Covering $|w - s_{\mathrm{ref}}| = 3$ by 256 overlapping complex interval arcs in 384-bit `FLINT/Arb` arithmetic certifies $M_3 := \sup_{|w-s_{\mathrm{ref}}|=3} |\tilde{\xi}_{\mathrm{DH}}(w)| \leq 5.934 \times 10^8$ (Appendix B), so $|c_k| \leq M_3 \cdot 3^{-k}$. Along the branch from $h = 0$ to $h = \frac{5}{8}$, every evaluation box $B_k \times H_k$ ($B_k := \{z : |\operatorname{Re}(z - z_k)| \leq r_k,\ |\operatorname{Im}(z - z_k)| \leq r_k\}$, $\sup_{z \in B_k} |z - z_k| = \sqrt{2}\, r_k$) satisfies $r_{\max} := \sup_{z \in B_k} |s(z) - s_{\mathrm{ref}}| + \sup |H_k| < \frac{6}{5}$, giving truncation ratio $q := r_{\max}/3 < \frac{2}{5}$. Partitioning $[0, \frac{5}{8}]$ into 360 overlapping interval balls $H_k = [h_k \pm \eta_k]$ (120 balls on $[0, 0.55]$ with $r_k = 8 \times 10^{-3}$, 100 balls on $[0.55, 0.620]$ with $r_k = 3 \times 10^{-3}$, 80 balls on $[0.620, 0.6247]$ with $r_k = 8 \times 10^{-4}$, and 60 balls on $[0.6247, \frac{5}{8}]$ with $r_k = 2 \times 10^{-4}$, with $0 \in H_1$ and $\frac{5}{8} \in H_{360}$), our 384-bit verification script truncates (3.5) and its z- and h-derivatives at order $K = 52$ (with analytical Cauchy remainders $\leq M_3 q^{52}/(1-q)$ and $\leq \frac{52 M_3}{3} q^{51}/(1-q)^2$) and verifies across all 360 slices that

$$N(B_k, H_k) := z_k - \frac{g_{\mathrm{sinc}}(z_k, h_k) + (H_k - h_k)\,\partial_h g_{\mathrm{sinc}}(z_k, H_k)}{\partial_z g_{\mathrm{sinc}}(B_k, H_k)} \subset \operatorname{int}(B_k),$$
$$\operatorname{Im} N(B_k, H_k) < 0,$$

together with $B(-2\mathrm{i}(\rho_{44838} - \frac{1}{2}), 10^{-12}) \subset B_1$ at $h = 0$ and $N(B_{k-1}, h_{k-1,k}) \subset B_k$ at each overlap midpoint $h_{k-1,k} \in H_{k-1} \cap H_k$ ($2 \leq k \leq 360$), gluing the slices into a single continuous non-real zero trajectory across $[0, \frac{5}{8}]$. At $h = \frac{5}{8}$ (and likewise at $h = \frac{625121}{10^6}$), evaluating the order-128 antiderivative difference $(2h)^{-1} \sum_{k=0}^{127} \frac{c_k}{k+1}[(u_+)^{k+1} - (u_-)^{k+1}]$ ($u_\pm := s(z) - s_{\mathrm{ref}} \pm h$, remainder $\leq M_3 q^{128}/(1-q) < 10^{-48}$) and $\partial_z g_{\mathrm{sinc}}(B_{5/8}, \frac{5}{8}) = \frac{\mathrm{i}}{4h}[\tilde{\xi}_{\mathrm{DH}}(s(B_{5/8}) +$

$h) - \tilde{\xi}_{\mathrm{DH}}(s(B_{5/8}) - h)]$ certifies $B_{5/8} \subset B_{360}$, $N(B_{5/8}) \subset \mathrm{int}(B_{5/8})$, and $\mathrm{Im}\, N(B_{5/8}) < 0$, proving (3.4) and $h^*_{\mathrm{global}}(G_{\mathrm{sinc}}; \Xi_{\mathrm{DH}}) > \frac{625121}{10^6} > \frac{5}{8}$ (Figure 1; similarly ρ_{2442} persists past $h = \frac{311}{500}$ up to $h_{\mathrm{col}}(\rho_{2442}) \approx 0.62453$).

(ii) Although $\{G_{\mathrm{sinc},h}\}_{h>0}$ is not a one-parameter semigroup in h, splitting $[-2h_1, 2h_1] = [-2h_1, 0] \cup [0, 2h_1]$ in $G_{\mathrm{sinc},2h_1}(-\partial_z^2)\Xi_{\mathrm{DH}}(z) = \frac{1}{4h_1}\int_{-2h_1}^{2h_1} \Xi_{\mathrm{DH}}(z + 2\mathrm{i}v)\,\mathrm{d}v$ and substituting $v = \mp h_1 + u$ ($u \in [-h_1, h_1]$) yields for every $h_1 > 0$ and all $z \in \mathbb{C}$ the exact doubling identity

$$G_{\mathrm{sinc},2h_1}(-\partial_z^2)\Xi_{\mathrm{DH}}(z) = \tfrac{1}{2}[F_{h_1}(z - 2\mathrm{i}h_1) + F_{h_1}(z + 2\mathrm{i}h_1)] = G_{\cosh,h_1}(-\partial_z^2)F_{h_1}(z), \tag{3.6}$$

where $F_{h_1}(z) := G_{\mathrm{sinc},h_1}(-\partial_z^2)\Xi_{\mathrm{DH}}(z) \not\equiv 0$. Because F_{h_1} is real-even and satisfies $|F_{h_1}(z)| \ll_{h_1} \mathrm{e}^{C|z|\log(|z|+2)}$, the entire function $w \mapsto F_{h_1}(\sqrt{w})$ has order $\frac{1}{2} < 1$ and Hadamard genus 0, so $F_{h_1}(z)$ has genus at most 1. By Lemma 3.1 and Lemma 2.1(i), all zeros of F_{h_1} lie in $|\mathrm{Im}\, z| \le 2(\sigma^* - \frac{1}{2})$. For every $h \ge 2\sigma^* - 1$, setting $h_1 := \frac{h}{2} \ge \sigma^* - \frac{1}{2}$ and applying Lemma 2.1(ii) to (3.6) shows that all zeros of $G_{\mathrm{sinc},h}(-\partial_z^2)\Xi_{\mathrm{DH}}(z)$ are real (and simple when $h > 2\sigma^* - 1$). □

We next resolve the critical boundary $h_* := \sigma^* - \frac{1}{2}$ ($\lim_{u\to\infty} S_{G_{\mathrm{sinc},h_*}}(u) = 2\sigma^* - 1$, Remark 2.8(b)) conditionally on standard L-function hypotheses.

Proposition 3.3 (Conditional resolution at the critical half-width $h_* = \sigma^* - \frac{1}{2}$)**.** *Assume the Riemann Hypothesis for $\zeta(s)$ and the Generalized Riemann Hypothesis for $L(s, \chi_{5,\mathrm{quad}})$, where $\chi_{5,\mathrm{quad}}$ is the real primitive Dirichlet character modulo 5. Then at $h_* := \sigma^* - \frac{1}{2}$, the entire function $F_*(z) := G_{\mathrm{sinc},h_*}(-\partial_z^2)\Xi_{\mathrm{DH}}(z)$ satisfies*

$$8 \le N_{\mathrm{off}}(G_{\mathrm{sinc},\sigma^*-1/2}(-\partial_z^2)\Xi_{\mathrm{DH}};\, \infty) < \infty, \tag{3.7}$$

where the lower bound $N_{\mathrm{off}}(F_*; \infty) \ge 8$ holds unconditionally.

Proof. Using the 320-bit interval enclosure $H_* = [\sigma^* - \frac{1}{2}]$ of radius 10^{-30} from Lemma 3.1, our verification script certifies disjoint non-real Interval–Newton boxes of radius 10^{-27} for $F_*(z)$ in $\{z : \mathrm{Re}\, z > 0,\ \mathrm{Im}\, z < 0\}$ near ρ_{2442} ($\mathrm{Re}\, z \approx 4884.89$, $\mathrm{Im}\, z \approx -0.0719$) and near ρ_{44838} ($\mathrm{Re}\, z \approx 89676.90$, $\mathrm{Im}\, z \approx -0.0663$); under $\{\pm z, \pm\bar{z}\}$, these yield at least 8 distinct non-real zeros unconditionally.

Assume now RH for $\zeta(s)$ and GRH for $L(s, \chi_{5,\mathrm{quad}})$. We first prove the pointwise bound $|\mathcal{D}_{\mathrm{DH}}(\sigma + \mathrm{i}\gamma)| \gg (\log\gamma)^{-1/3}$ for all $\gamma \ge \gamma_0$ and all $\sigma \ge \sigma^* - (\log\gamma)^{-1/3}$. For $\sigma > 3$, $|\mathcal{D}_{\mathrm{DH}}(\sigma + \mathrm{i}\gamma)| \ge 2 - \zeta(3) > \frac{3}{4}$. On $\sigma \ge 1.07$, write $\mathcal{D}_{\mathrm{DH}}(w) = \frac{1+\mathrm{i}\kappa}{2}L(w, \chi_5)(\mathrm{e}^{\mathcal{L}(w)} - \mathrm{e}^{\mathrm{i}\theta_\kappa})$, where $\theta_\kappa := \pi - 2\arctan\kappa = 2.5880\ldots \in (\frac{\pi}{2}, \pi)$ (by (3.3)) and $\mathcal{L}(w) := \sum_{p\equiv 2,3 \pmod 5} \log\frac{1-\chi_5(p)p^{-w}}{1+\chi_5(p)p^{-w}} =: R(w) + \mathrm{i}\Phi(w)$ uses the principal branch. Euler-product comparison gives $|L(w, \chi_5)| \ge (1 + 5^{-1.07})\frac{\zeta(2.14)}{\zeta(1.07)} > \frac{3}{25}$ on $\sigma \ge 1.07$, while $2\arctan x < \log\frac{1+x}{1-x}$ ($0 < x < 1$) gives $|\Phi(w)| \le \sum_{p\equiv 2,3} 2\arctan(p^{-1.07}) < \log\frac{(1-5^{-1.07})\zeta(1.07)}{L(1.07,\chi_{5,\mathrm{quad}})} < 3.30 < 2\pi - \theta_\kappa - \frac{1}{3}$, so $\Phi(w) - \theta_\kappa \in [-2\pi + \frac{1}{3}, \frac{\pi}{2}]$ and hence $|\mathrm{e}^{\mathcal{L}(w)} - \mathrm{e}^{\mathrm{i}\theta_\kappa}| \ge \min(\frac{2}{\pi}|\theta_\kappa - \Phi(w)|, \sin\frac{1}{3})$.

At $w = \sigma^* + \mathrm{i}\gamma$, write $r_p := p^{-\sigma^*} < \frac{1}{2}$ and $\chi_5(p)p^{-\sigma^*-\mathrm{i}\gamma} = -\mathrm{i}r_p\mathrm{e}^{\mathrm{i}\phi_p(\gamma)}$ for $p \equiv 2, 3 \pmod 5$. Since $\chi_5(p)^2 = -1$, squaring gives $\mathrm{e}^{2\mathrm{i}\phi_p(\gamma)} = \mathrm{e}^{-2\mathrm{i}\gamma\log p}$, so $1 - \cos(2\gamma\log p) = 2(1 - \cos\phi_p)(1 + \cos\phi_p) \le 4(1 - \cos\phi_p)$. By the Mean Value Theorem on $2\arctan r_p - \mathrm{Im}\log\frac{1+\mathrm{i}r_p\mathrm{e}^{\mathrm{i}\phi_p}}{1-\mathrm{i}r_p\mathrm{e}^{\mathrm{i}\phi_p}} = \arctan\frac{2r_p}{1-r_p^2} - \arctan\frac{2r_p\cos\phi_p}{1-r_p^2}$, each prime $p \equiv 2, 3 \pmod 5$ contributes at least $\frac{2(1-r_p^2)}{(1+r_p^2)^2}r_p(1 - \cos\phi_p) \ge \frac{6}{25}p^{-\sigma^*}(1 - \cos(2\gamma\log p)) \ge 0$ to $\theta_\kappa - \Phi(\sigma^* + \mathrm{i}\gamma)$. Introduce $k(u) := \mathrm{e}^{-u/2} - \mathrm{e}^{-u} \ge 0$ ($u \ge 0$), whose Mellin transform $\widehat{k}(w) = (2^w - 1)\Gamma(w)$ is holomorphic on $\mathrm{Re}\, w > -1$, and set

$$f_5(s) := \frac{1}{2}\left[-\frac{\zeta'}{\zeta}(s) + \frac{L'}{L}(s, \chi_{5,\mathrm{quad}}) - \frac{\log 5}{5^s - 1}\right] = \sum_{p\equiv 2,3 \pmod 5} \sum_{\substack{m\ge 1 \\ m \text{ odd}}} (\log p)\, p^{-ms}.$$

Using $1-\cos(2m\gamma\log p)\le m^2(1-\cos(2\gamma\log p))$, $(\log p)k(p/X)\le\log(2X)$ for $X\ge \mathrm{e}^2$, and the prime-power tail $\sum_{\ell=1}^{\infty}(\log p)(2\ell+1)^2p^{-2\ell\sigma^*}<4$, the smoothed sum

$$S_\Lambda(X,\gamma):=\sum_{p\equiv 2,3 \pmod 5}\ \sum_{\substack{m\ge 1\\ m\ \text{odd}}}(\log p)\,p^{-m\sigma^*}\bigl(1-\cos(2m\gamma\log p)\bigr)k(p^m/X)$$

satisfies $S_\Lambda(X,\gamma)\le\frac{50}{3}(\log X)(\theta_\kappa-\Phi(\sigma^*+\mathrm{i}\gamma))$. Expressing $S_\Lambda(X,\gamma)=\mathrm{Re}[I_5(X,0)-I_5(X,2\gamma)]$ via

$$I_5(X,t):=\frac{1}{2\pi\mathrm{i}}\int_{1-\mathrm{i}\infty}^{1+\mathrm{i}\infty}f_5(\sigma^*+\mathrm{i}t+w)\,\widehat{k}(w)X^w\,\mathrm{d}w$$

and shifting to $\mathrm{Re}\,w=\frac12-\sigma^*+\frac{1}{100}>-1$ under RH and GRH via $|f_5(\frac{51}{100}+\mathrm{i}\tau)|\ll\log(|\tau|+2)$ [29, Thm. 14.5], [19, Thm. 13.13] picks up the residue $\frac12 c_*X^{-(\sigma^*-1)}$ at $w=1-\sigma^*$ ($c_*:=(2^{1-\sigma^*}-1)\Gamma(1-\sigma^*)>\frac{7}{10}$) plus $\mathcal{O}(\mathrm{e}^{-\pi\gamma}X^{-(\sigma^*-1)})$ at $w=1-\sigma^*-2\mathrm{i}\gamma$, yielding $S_\Lambda(X,\gamma)=\frac{c_*}{2}X^{-(\sigma^*-1)}+\mathcal{O}(X^{-1/2-(\sigma^*-1)+1/100}\log\gamma)$. Setting $X:=(\log\gamma)^{21/10}$ (since $\frac{21}{10}(\sigma^*-1)<\frac{26}{100}$) gives $\theta_\kappa-\Phi(\sigma^*+\mathrm{i}\gamma)\gg(\log\gamma)^{-3/10}$. Because $\sup_{\sigma\ge1.08}|\partial_\sigma\mathcal{L}(\sigma+\mathrm{i}\gamma)|<\infty$ and $(\log\gamma)^{-1/3}=o((\log\gamma)^{-3/10})$, we obtain $\theta_\kappa-\Phi(\sigma+\mathrm{i}\gamma)\gg(\log\gamma)^{-3/10}$ on $|\sigma-\sigma^*|\le(\log\gamma)^{-1/3}$, while on $\sigma\in[\sigma^*+(\log\gamma)^{-1/3},3]$ monotonicity gives $\theta_\kappa-\Phi(\sigma+\mathrm{i}\gamma)\ge2(\arctan(2^{-\sigma^*})-\arctan(2^{-\sigma}))\gg(\log\gamma)^{-1/3}$. Thus $|\mathcal{D}_{\mathrm{DH}}(\sigma+\mathrm{i}\gamma)|\gg(\log\gamma)^{-1/3}$ for all $\sigma\ge\sigma^*-(\log\gamma)^{-1/3}$, $\gamma\ge\gamma_0$.

To deduce $N_{\mathrm{off}}(F_*;\infty)<\infty$, set $\eta:=\frac{1}{64}$ and apply Lemma 2.5 to G_{sinc,h_*} ($\tau_0=2h_*=2\sigma^*-1$, $h_G(w)=\frac{\sinh(h_*w)}{h_*w}$). Since $\frac{h_G(w-u)}{h_G(w)}=\frac{w}{w-u}\mathrm{e}^{-h_*u}\frac{1-\mathrm{e}^{-2h_*(w-u)}}{1-\mathrm{e}^{-2h_*w}}$, splitting $1\le n\le N_\gamma$ in (2.6) at $\lfloor\exp((\log\gamma)^{1/2})\rfloor$ strengthens (2.5) on $\delta\in[-2\eta,h_*]$ ($\sigma_z+h_*=\sigma^*+\delta>1.08>1+3\eta$) to $|r_+(z)|\ll(\log\gamma)^{-1}$. Combining $|r_+(z)|\ll(\log\gamma)^{-1}$ with $|\mathcal{D}_{\mathrm{DH}}(s(z)+h_*)|\gg(\log\gamma)^{-1/3}$ gives $I_+(z)=\mathcal{M}_*(z)(1+\mathcal{O}((\log\gamma)^{-2/3}))$ uniformly on $\delta\in[-(\log\gamma)^{-1/3},h_*]$, where $\mathcal{M}_*(z):=\Gamma_{\mathrm{DH}}(s(z))h_G(\Omega(z))\mathcal{D}_{\mathrm{DH}}(s(z)+h_*)\neq0$. For $\delta\in[\eta,h_*]$, Lemma 2.4(i) gives $|F_*(z)-I_+(z)|\ll\gamma^{-5\eta/32}(\log\gamma)^{1/3}|\mathcal{M}_*(z)|$; for $\delta\in[\frac12(\log\gamma)^{-1/3},\eta]$, $|I_-(z)|=|I_+(\bar z)|\ll\gamma^{-\delta}(\log\gamma)^{1/3}|\mathcal{M}_*(z)|\le\exp(-\frac12(\log\gamma)^{2/3}+\frac13\log\log\gamma)|\mathcal{M}_*(z)|\to0$. Finally, on $|\delta|\le\frac12(\log\gamma)^{-1/3}$, both z and $\bar z$ satisfy the cap asymptotic, so $F_*(z)=\mathcal{M}_*(z)+\overline{\mathcal{M}_*(\bar z)}+\mathcal{O}((\log\gamma)^{-2/3})\max(|\mathcal{M}_*(z)|,|\overline{\mathcal{M}_*(\bar z)}|)$, while $|\mathcal{D}'_{\mathrm{DH}}|\le-\zeta'(1.08)$ gives $|\frac{\mathrm{d}}{\mathrm{d}z}\log\mathcal{D}_{\mathrm{DH}}(s(z)+h_*)|\ll(\log\gamma)^{1/3}$ and replaces (2.10) by $\frac{\mathrm{d}}{\mathrm{d}z}\log\mathcal{M}_*(z)=-\frac{\pi}{8}+\frac{\mathrm{i}}{4}\log\frac{5\gamma}{2\pi}+\mathcal{O}((\log\gamma)^{1/3})$. Because $(\log\gamma)^{1/3}=o(\log\gamma)$ and $\delta_k(\log x_k)^{1/3}\ll(\log x_k)^{-2/3}\to0$ on the vertical edges of the Rouché rectangles $\mathcal{R}_k=[x_k,x_{k+1}]\times[-2\mathrm{i}\delta_k,2\mathrm{i}\delta_k]$ ($\delta_k=\frac{4}{\log(x_k/2)}$), items (a)–(b) of Case 2 in Theorem 2.6 carry over verbatim, proving $N_{\mathrm{off}}(F_*;\infty)<\infty$. □

Remark 3.4 (Unconditional Vinogradov–Korobov and Baker–Wüstholz barriers). Replacing RH and GRH in Proposition 3.3 by the unconditional Vinogradov–Korobov zero-free region $1-c(\log(|\tau|+3))^{-2/3}(\log\log(|\tau|+3))^{-1/3}$ [15] requires Mellin scale $\log X\asymp(\log\gamma)^{2/3}(\log\log\gamma)^{4/3}$ to beat the vertical-line contour integral and yields only $\theta_\kappa-\Phi(\sigma^*+\mathrm{i}\gamma)\gg\exp(-c'(\log\gamma)^{2/3}(\log\log\gamma)^{4/3})$, which decays faster than the Borel cap error $\mathcal{O}((\log\gamma)^{-1})$ in Lemma 2.5. Likewise, bounding the simultaneous Diophantine defect $\sum_{p\le P}p^{-\sigma^*}(1-\cos(2\gamma\log p))$ directly via linear forms in logarithms [1] requires truncating the prime tail $\sum_{p>P}p^{-\sigma^*}\asymp\frac{P^{-(\sigma^*-1)}}{(\sigma^*-1)\log P}$ below $(\log\gamma)^{-1}$, forcing $\pi(P)\asymp(\log\gamma)^{\frac{1}{\sigma^*-1}}(\log\log\gamma)^{-\frac{\sigma^*}{\sigma^*-1}}=(\log\gamma)^{8.308\ldots-o(1)}$ independent logarithms whose n^n dependence produces an even weaker doubly exponential bound.

4. Gauss–Weierstrass Asymptotics and the Ki–Kim–Lee Upper Bound

In this section we adapt the method of Ki, Kim, and Lee [14] (who proved $\Lambda_\zeta<1/2$ by combining large-height zero simplicity of H_t with de Bruijn's hyperbola inequality [9]) to $\Xi_{\mathrm{DH}}(z)$. Writing the Gaussian convolution $H_t^{\mathrm{DH}}(z)=(4\pi t)^{-1/2}\int_{-\infty}^{\infty}\mathrm{e}^{-w^2/(4t)}\Xi_{\mathrm{DH}}(z-\mathrm{i}w)\,\mathrm{d}w$

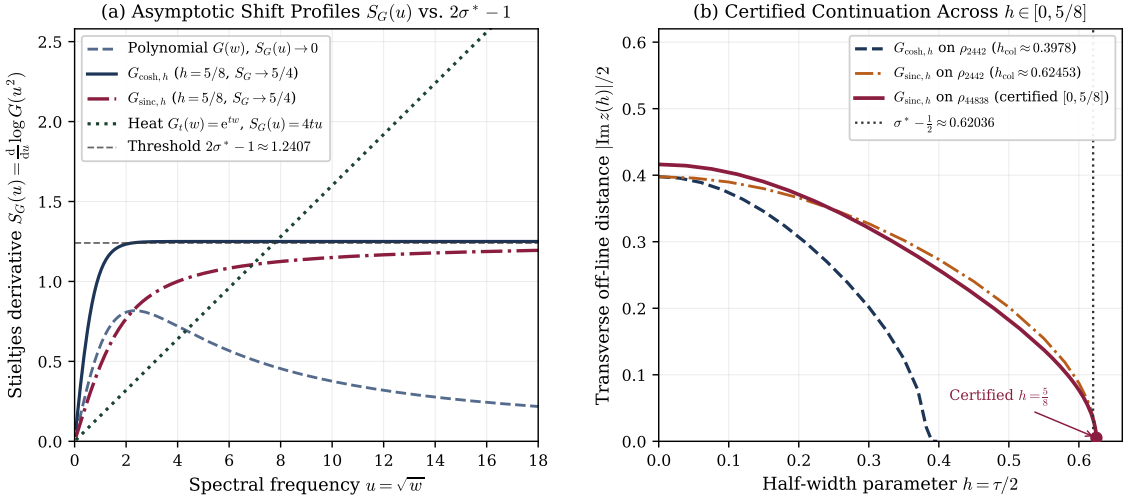


FIGURE 1. (a) Stieltjes logarithmic-derivative profiles $S_G(u)$ relative to $2\sigma^*-1$. (b) Transverse off-line ordinate $|\operatorname{Im} z(h)|/2$ comparing $G_{\cosh,h}$ on ρ_{2442} with $G_{\mathrm{sinc},h}$ on ρ_{2442} ($h_{\mathrm{col}}\approx 0.62453$) and on ρ_{44838} (360-ball singularity-free continuation across $h\in[0,5/8]$, $h_{\mathrm{col}}\approx 0.62512$).

($t>0$) at $z=2\gamma-2\mathrm{i}\delta$ via $v=\delta+w/2$, splitting $\mathbb{R}=[0,\infty)\cup(-\infty,0)$, and applying $\xi_{\mathrm{DH}}(\frac{1}{2}-v+\mathrm{i}\gamma)=\overline{\xi_{\mathrm{DH}}(\frac{1}{2}+v+\mathrm{i}\gamma)}$ for real γ,v gives

$$\begin{aligned} H_t^{\mathrm{DH}}(2\gamma-2\mathrm{i}\delta) &= A_t(\gamma,\delta)+\overline{A_t(\gamma,-\delta)},\\ A_t(\gamma,\delta) &:= \frac{1}{\sqrt{\pi t}}\int_0^\infty \mathrm{e}^{-(v-\delta)^2/t}\,\xi_{\mathrm{DH}}\Big(\tfrac{1}{2}+v+\mathrm{i}\gamma\Big)\,\mathrm{d}v,\\ \partial_\gamma A_t(\gamma,\delta) &= \mathrm{i}\,\partial_\delta A_t(\gamma,\delta)-\frac{\mathrm{i}}{\sqrt{\pi t}}\,\mathrm{e}^{-\delta^2/t}\,\xi_{\mathrm{DH}}\Big(\tfrac{1}{2}+\mathrm{i}\gamma\Big), \end{aligned} \tag{4.1}$$

where the third relation follows on integrating $\partial_\delta(\mathrm{e}^{-(v-\delta)^2/t})=-\partial_v(\mathrm{e}^{-(v-\delta)^2/t})$ by parts over $[0,\infty)$, the boundary contribution at $+\infty$ vanishing because $|\xi_{\mathrm{DH}}(\frac{1}{2}+v+\mathrm{i}\gamma)|\ll_\gamma (1+v)^C\mathrm{e}^{\frac{v}{2}\log(v+2)}$ is dominated by $\mathrm{e}^{-(v-\delta)^2/t}$.

Lemma 4.1 (de Bruijn [9, Thms. 12 and 13]). *Let $F(z)$ be a real-even entire function of genus 0 or 1 whose zeros all satisfy $|\operatorname{Im} z|\le\Delta_0$.*

(i) *For every $t>0$, all zeros of $\mathrm{e}^{-t\partial_z^2}F(z)$ satisfy $|\operatorname{Im} z|\le\sqrt{\max(0,\Delta_0^2-2t)}$; in particular, $\Lambda_{\mathrm{DH}}\le 2(\sigma^*-\frac{1}{2})^2$.*

(ii) *If $x_*\in\mathbb{R}$ and $\widetilde{\Delta}\ge 0$ are such that every zero z_ρ of F satisfies $(\operatorname{Im} z_\rho)^2-(\operatorname{Re} z_\rho-x_*)^2\le\widetilde{\Delta}^2$, then every zero z of $\mathrm{e}^{-t\partial_z^2}F(z)$ on the vertical line $\operatorname{Re} z=x_*$ satisfies $(\operatorname{Im} z)^2\le\max(0,\widetilde{\Delta}^2-2t)$.*

We next state the explicit Dirichlet bounds proved in Appendix A.

Lemma 4.2 (Explicit Abel and Dirichlet-tail bounds). *For all $v\ge 0$, $|\mathcal{D}_{\mathrm{DH}}(\frac{1}{2}+v+\mathrm{i}\tau)|\le 4|\tau|^{1/2}$ when $|\tau|\ge 100$, and $|\mathcal{D}_{\mathrm{DH}}(\frac{1}{2}+v+\mathrm{i}\tau)|\le 38$ when $|\tau|\le 100$. Moreover, for all $\tau\in\mathbb{R}$ and $u\ge\frac{5}{4}$, $\sum_{n=2}^\infty|a_{\mathrm{DH}}(n)|n^{-1/2-u}<\frac{1}{2}$, so $\operatorname{Re}\mathcal{D}_{\mathrm{DH}}(\frac{1}{2}+u+\mathrm{i}\tau)>\frac{1}{2}$ and $|\mathcal{D}_{\mathrm{DH}}(\frac{1}{2}+u+\mathrm{i}\tau)|<\frac{3}{2}$.*

Proposition 4.3 (Explicit height cutoff for H_t^{DH}; cf. Ki–Kim–Lee [14]). *Fix $t>0$ and set $\Delta:=\sigma^*-\frac{1}{2}<\frac{5}{8}$. Define the explicit cutoff height*

$$\gamma_0(t):=\max\Big(400+64t^3,\ \frac{2\pi}{5}\exp\Big(\frac{4}{t}\Big(\Delta+5+6\sqrt{t}+2t\Big)\Big)\Big). \tag{4.2}$$

Then for all $|\gamma| \geq \gamma_0(t)$, $H_t^{\mathrm{DH}}(2\gamma - 2\mathrm{i}\delta) \neq 0$ for every $\delta \neq 0$, and every real zero of $H_t^{\mathrm{DH}}(x)$ in $|x| \geq 2\gamma_0(t)$ is simple.

Proof. By evenness and Lemma 4.1(i), it suffices to consider $\gamma \geq \gamma_0(t)$ and $|\delta| \leq \Delta$. In (4.1), fix any $\eta \in [-\Delta, \Delta]$ and deform $v \in [0, \infty)$ to $\mathcal{V}_\theta \cup \mathcal{C}_\theta$, where $\mathcal{V}_\theta := \{\mathrm{i}y : 0 \leq y \leq \theta\}$, $\mathcal{C}_\theta := \{u + \mathrm{i}\theta : u \geq 0\}$, and $\theta := \frac{\pi t}{8}$; Cauchy's theorem applies because ξ_{DH} is entire and $|\mathrm{e}^{-(u+\mathrm{i}y-\eta)^2/t}\xi_{\mathrm{DH}}(\frac{1}{2} + u + \mathrm{i}(\gamma + y))| \ll_{\gamma,\theta} \mathrm{e}^{-u^2/(2t)} \to 0$ as $u \to +\infty$ uniformly for $y \in [0, \theta]$. Set $\tilde{\gamma} := \gamma + \theta > \gamma$ and $K_0(\gamma, \eta) := \mathrm{e}^{\frac{\pi^2 t}{64} + \mathrm{i}\frac{\pi}{4}\eta}\Gamma_{\mathrm{DH}}(\frac{1}{2} + \mathrm{i}\tilde{\gamma}) \neq 0$. Along $v = u + \mathrm{i}\theta$, expanding $-(u + \mathrm{i}\theta - \eta)^2/t = -\frac{(u-\eta)^2}{t} + \frac{\pi^2 t}{64} - \mathrm{i}\frac{\pi}{4}(u - \eta)$ cancels the $+\mathrm{i}\frac{\pi}{4}u$ Stirling phase of $\Gamma_{\mathrm{DH}}(\frac{1}{2} + u + \mathrm{i}\tilde{\gamma})/\Gamma_{\mathrm{DH}}(\frac{1}{2} + \mathrm{i}\tilde{\gamma}) = (\frac{5\tilde{\gamma}}{2\pi})^{u/2}\mathrm{e}^{\mathrm{i}\frac{\pi}{4}u + E_{\mathrm{St}}(u,\tilde{\gamma})}$, giving

$$\frac{\sqrt{\pi t}\,\mathrm{e}^{-u_0^2/t + \eta^2/t}}{K_0(\gamma, \eta)}\, A_t(\gamma, \eta) = \int_0^\infty \mathrm{e}^{-(u-u_0)^2/t + E_{\mathrm{St}}(u,\tilde{\gamma})}\, \mathcal{D}_{\mathrm{DH}}\Big(\tfrac{1}{2} + u + \mathrm{i}\tilde{\gamma}\Big)\, \mathrm{d}u + \mathcal{E}_{\mathcal{V}}(\gamma, \eta), \tag{4.3}$$

where $u_0 = u_0(\gamma, \eta) := \eta + \frac{t}{4}\log\frac{5\tilde{\gamma}}{2\pi} > 5 + 6\sqrt{t} + 2t$. As shown in Appendix A, $|E_{\mathrm{St}}(u, \tilde{\gamma})| \leq \frac{u^2+6u}{4\tilde{\gamma}}$ for all $u \geq 0$, so $|E_{\mathrm{St}}(u, \tilde{\gamma})| < \frac{1}{10}$ on $|u - u_0| \leq 3\sqrt{t}$ (where $u > 5 > \frac{5}{4}$ and $\mathrm{Re}(\mathrm{e}^{E_{\mathrm{St}}}\mathcal{D}_{\mathrm{DH}}) \geq \mathrm{e}^{-1/10}(\frac{1}{2}\cos\frac{1}{10} - \frac{3}{2}\sin\frac{1}{10}) > 0.31$), while $[0, \frac{5}{4}] \cup [\frac{5}{4}, u_0 - 3\sqrt{t}]$, $\mathcal{V}_\theta$, and $u \geq u_0 + 3\sqrt{t}$ contribute less than $0.015\sqrt{\pi t}$ in modulus, leaving real part $> 0.29\sqrt{\pi t}$. Writing $\frac{2}{t}(v - \eta) = (\frac{1}{2}\log\frac{5\tilde{\gamma}}{2\pi} + \mathrm{i}\frac{\pi}{4}) + \frac{2}{t}(u - u_0)$ in $\partial_\eta A_t(\gamma, \eta)$ bounds the centered-moment remainder by $\frac{6.7}{\sqrt{t}} < \frac{1}{4}\log\frac{5\tilde{\gamma}}{2\pi}$ (Appendix A), whence $A_t(\gamma, \eta) \neq 0$ and

$$\mathrm{Re}\Big(\frac{\partial_\eta A_t(\gamma, \eta)}{A_t(\gamma, \eta)}\Big) \geq \frac{1}{4}\log\frac{5\tilde{\gamma}}{2\pi} > 1 \qquad \forall\, \eta \in [-\Delta, \Delta]. \tag{4.4}$$

Integrating (4.4) over $[-\delta, \delta]$ for $\delta \in (0, \Delta]$ gives $|A_t(\gamma, \delta)| > |A_t(\gamma, -\delta)|$ and hence $H_t^{\mathrm{DH}}(2\gamma - 2\mathrm{i}\delta) \neq 0$. At $\delta = 0$, $H_t^{\mathrm{DH}}(2\gamma) = 2|A_t(\gamma, 0)|\cos\Theta_t(\gamma)$ with $\Theta_t(\gamma) := \arg A_t(\gamma, 0)$, and the third identity of (4.1) together with the boundary ratio bound $< 10^{-2}$ from Appendix A gives

$$\Theta_t'(\gamma) = \mathrm{Re}\Big(\frac{\partial_\delta A_t(\gamma, 0)}{A_t(\gamma, 0)}\Big) - \mathrm{Re}\Big(\frac{\xi_{\mathrm{DH}}(\frac{1}{2} + \mathrm{i}\gamma)}{\sqrt{\pi t}\, A_t(\gamma, 0)}\Big) \geq \frac{1}{4}\log\frac{5\gamma}{2\pi} - 10^{-2} > 1 > 0, \tag{4.5}$$

so $\frac{\mathrm{d}}{\mathrm{d}\gamma}H_t^{\mathrm{DH}}(2\gamma) = \mp 2|A_t(\gamma, 0)|\Theta_t'(\gamma) \neq 0$ at every real zero $\cos\Theta_t(\gamma) = 0$. □

Applying the finite-box hyperbola argument of Ki, Kim, and Lee [14, §3] and de Bruijn [9, Thm. 12] immediately sharpens Lemma 4.1(i) to a strict inequality.

Corollary 4.4 (Strict upper bound for Λ_{DH}; after Ki–Kim–Lee [14])**.** *The de Bruijn–Newman constant of $\Xi_{\mathrm{DH}}(z)$ satisfies*

$$\Lambda_{\mathrm{DH}} < 2\Big(\sigma^* - \tfrac{1}{2}\Big)^2 = 0.76969925830451198120\ldots \tag{4.6}$$

Proof. Set $\Delta_0 := 2(\sigma^* - \frac{1}{2})$ and $t_1 := \frac{1}{2} < \frac{1}{2}\Delta_0^2 = 2(\sigma^* - \frac{1}{2})^2$. By Proposition 4.3, $H_{t_1}^{\mathrm{DH}}(z)$ has only real zeros in $|\mathrm{Re}\, z| \geq 2\gamma_0(t_1)$ and hence at most finitely many non-real zeros overall. If $H_{t_1}^{\mathrm{DH}}$ has no non-real zeros, then $\Lambda_{\mathrm{DH}} \leq t_1 < 2(\sigma^* - \frac{1}{2})^2$. Otherwise, $Y_1 := \max\{|\mathrm{Im}\, w| : H_{t_1}^{\mathrm{DH}}(w) = 0\} > 0$ is attained at some $w_* = x_* + \mathrm{i}y_*$ with $|x_*| < 2\gamma_0(t_1)$. By Lemma 3.1, every zero z_ρ of $\Xi_{\mathrm{DH}}(z)$ satisfies $|\mathrm{Im}\, z_\rho| < \Delta_0$ strictly. Setting $R := 2\gamma_0(t_1) + \Delta_0 + 1$, the finitely many zeros of Ξ_{DH} in $|\mathrm{Re}\, z_\rho| \leq R$ satisfy $\Delta_R := \max_{|\mathrm{Re}\, z_\rho| \leq R}|\mathrm{Im}\, z_\rho| < \Delta_0$, while zeros with $|\mathrm{Re}\, z_\rho| > R$ satisfy $|\mathrm{Re}\, z_\rho - x_*| > \Delta_0 + 1 > |\mathrm{Im}\, z_\rho|$, so $(\mathrm{Im}\, z_\rho)^2 - (\mathrm{Re}\, z_\rho - x_*)^2 \leq \Delta_R^2 < \Delta_0^2$ globally. Lemma 4.1(ii) at $\mathrm{Re}\, z = x_*$ yields $Y_1^2 \leq \max(0, \Delta_R^2 - 2t_1)$, and Lemma 4.1(i) applied to $H_{t_1 + Y_1^2/2}^{\mathrm{DH}} = \mathrm{e}^{-(Y_1^2/2)\partial_z^2}H_{t_1}^{\mathrm{DH}}$ gives $\Lambda_{\mathrm{DH}} \leq t_1 + \frac{1}{2}Y_1^2 \leq \max(t_1, \frac{1}{2}\Delta_R^2) < \frac{1}{2}\Delta_0^2 = 2(\sigma^* - \frac{1}{2})^2$. □

Remark 4.5 (Modified Dirichlet-polynomial ℓ^1 barrier versus Euler-product L-functions). For primitive real-valued Dirichlet characters of conductor $3 \leq m \leq 400$ (whose L-functions possess an Euler product and have all non-trivial zeros inside $0 < \mathrm{Re}\, s < 1$),

Freire et al. [11] recently established $\Lambda_L \le \frac{953}{3200} = 0.2978125$ by combining Polymath-type effective heat-flow approximations [24] with numerical GRH verifications and interval computations. For $\mathcal{D}_{\mathrm{DH}}(s)$, the absence of an Euler product prevents factoring out 1-line zeros; in Dobner's Riemann–Siegel-truncated modified Dirichlet polynomial $\sum_{n=1}^{N(\gamma)} a_{\mathrm{DH}}(n)\,\mathrm{e}^{\frac{t}{4}\log^2 n}\,n^{-\frac12-\frac{t}{4}\log\frac{5\gamma}{2\pi}-\mathrm{i}\gamma}$ ($N(\gamma) := \lfloor\sqrt{5\gamma/(2\pi)}\rfloor$; cf. (2.6) and [10, Lem. 2.3]), the non-multiplicative Gaussian weight $\mathrm{e}^{\frac{t}{4}\log^2 n}$ cancels half of the effective abscissa damping at $n \sim N(\gamma)$ and inflates the ℓ^1-dominance threshold $\sum_{n=2}^{N(\gamma)} |a_{\mathrm{DH}}(n)|\,\mathrm{e}^{\frac{t}{4}\log^2 n}\,n^{-\frac12-\frac{t}{4}\log\frac{5\gamma}{2\pi}} < 1$ from $\gamma_{\ell^1}(0.50) \approx 1.52\times10^4$ ($N = 110$) to $\gamma_{\ell^1}(0.25) \approx 4.18\times10^8$ ($N = 18{,}235$). Lowering Corollary 4.4 toward $\frac14$ therefore requires a finite-box argument-principle cover up to heights of order 10^8 unless oscillatory cancellation in the modified Dirichlet polynomial is exploited beyond ℓ^1.

5. Certified Lower Bound on Λ_{DH} via Ball Arithmetic

5.1. **Rigorous theta-kernel truncation and Interval–Newton certificate.** To certify an unconditional lower bound on Λ_{DH} free of quadrature aliasing or root-tracking heuristics, we enclose $H_t^{\mathrm{DH}}(z) = 2\int_0^\infty \mathrm{e}^{tu^2}\Phi_{\mathrm{DH}}(u)\cos(zu)\,\mathrm{d}u$ and its first three z-derivatives in complex ball arithmetic [12] on $[0,U]$ with analytical remainder bounds.

Lemma 5.1 (Theta-kernel truncation and fourth-derivative enclosure)**.** *Let $0 \le t \le \frac14$, $2 \le U \le \frac52$, $N \ge 25$, and $|y| \le 1$, where $y = \operatorname{Im} z$. Approximating $\Phi_{\mathrm{DH}}(u)$ on $[0,U]$ by $\Phi_{\mathrm{DH}}^{(<N)}(u) := 4\mathrm{e}^{3u}\sum_{n=1}^{N-1} n\,a_{\mathrm{DH}}(n)\,\mathrm{e}^{-\pi n^2\mathrm{e}^{4u}/5}$ incurs a truncation error bounded for $k \in \{0,1,2,3,4\}$ by*

$$\left|\partial_z^k H_t^{\mathrm{DH}}(z) - 2\int_0^U \mathrm{e}^{tu^2}\Phi_{\mathrm{DH}}^{(<N)}(u)\,\partial_z^k\cos(zu)\,\mathrm{d}u\right| \le E_1(U,N,k,|y|) + E_2(U), \tag{5.1}$$

where $r_N := \mathrm{e}^{-\pi N/5}$, $E_1(U,N,k,|y|) := 8U^{k+1}\mathrm{e}^{tU^2+(3+|y|)U}\frac{r_N^N(N-(N-1)r_N)}{(1-r_N)^2}$, and $E_2(U) := \frac{40}{\pi}\mathrm{e}^{-4U-\frac{\pi}{10}\mathrm{e}^{4U}}$. Moreover, on any closed box B with $\max_{z\in B}|\operatorname{Im} z| \le y_{\max} \le 1$, one has

$$\sup_{z\in B}\left|\partial_z^4 H_t^{\mathrm{DH}}(z)\right| \le 4N(N-1)U^5\mathrm{e}^{tU^2+3U}\cosh(Uy_{\max}) + E_1(U,N,4,y_{\max}) + E_2(U) =: M_4(B). \tag{5.2}$$

The proof of Lemma 5.1 is given in Appendix B. At $z_0 \approx 480.69 - 9.54\times10^{-7}\,\mathrm{i}$ ($\gamma \approx 240.35$), $|\partial_z H_t^{\mathrm{DH}}(z_0)| \approx 7\times10^{-88}$ due to $\mathrm{e}^{-\pi\gamma/4} \approx 10^{-82}$ oscillatory decay and proximity to coalescence; expanding $\partial_z H_t^{\mathrm{DH}}(z)$ on $B(z_0,r) := \{z\in\mathbb{C} : |\operatorname{Re}(z-z_0)| \le r,\ |\operatorname{Im}(z-z_0)| \le r\}$ to third order via $\partial_z H_t^{\mathrm{DH}}(z_0) + (z-z_0)\partial_z^2 H_t^{\mathrm{DH}}(z_0) + \frac12(z-z_0)^2\partial_z^3 H_t^{\mathrm{DH}}(z_0)$ with Taylor remainder bounded by $\frac16|z-z_0|^3 M_4(B) \le \frac{\sqrt2}{3}r^3 M_4(B)$ allows rigorous certification at sober 512-bit working precision with box radius $r = 10^{-32}$.

Theorem 5.2 (Interval-certified lower bound on Λ_{DH})**.** *At exact rational $t_* := \frac{28825920237}{125000000000} = 0.230607361896$, the entire function $H_{t_*}^{\mathrm{DH}}(z)$ possesses a certified simple non-real zero inside $B(z_{240}, 10^{-32})$ centered at*

$$\begin{aligned} z_{240} = {} & 480.690231042507472707203841348070155707 1 \\ & - 9.535301365553738949062115339251329328331\times10^{-7}\,\mathrm{i}, \end{aligned} \tag{5.3}$$

originating from the off-line zero $\rho_{240} = 0.8695305796\ldots + 240.4046723514\ldots\,\mathrm{i}$ of $\mathcal{D}_{\mathrm{DH}}(s)$. Consequently, $\Lambda_{\mathrm{DH}} > \frac{28825920237}{125000000000} = 0.230607361896$.

Proof. Evaluating $\partial_z^k H_{t_*}^{\mathrm{DH}}(z_{240})$ ($k\in\{0,1,2,3\}$) with $U = \frac{23}{10} = 2.30$ and $N = 34$ via 512-bit `FLINT/Arb` Gauss–Legendre ball quadrature (`acb.integral` [12]) plus the analytical remainder balls of Lemma 5.1, and enclosing $\partial_z H_{t_*}^{\mathrm{DH}}(B)$ on $B := B(z_{240}, 10^{-32})$ via (5.2), our verification script confirms that $0 \notin \partial_z H_{t_*}^{\mathrm{DH}}(B)$, $N(B) := z_{240} - H_{t_*}^{\mathrm{DH}}(z_{240})/\partial_z H_{t_*}^{\mathrm{DH}}(B) \subset \operatorname{int}(B)$, and $\sup_{w\in N(B)}\operatorname{Im} w < -9.53\times10^{-7} < 0$ (Table 1). By Brouwer's fixed-point theorem

TABLE 1. Summary of `FLINT/Arb` ball-arithmetic enclosures verified by `verify_paper15_debruijn_newman_dh.py`.

Target / Witness	Param.	Certified Midpoint / Cover (16 digits shown)	Radius r
Supremum σ^* (320-bit)	IVT	1.1203624981833251	10^{-30}
G_{sinc} cover (384-bit)	$[0, \frac{5}{8}]$	360 overlapping balls H_k (ρ_{44838}), $\mathrm{Im}\, N(B_k, H_k) < 0$	$\le 8 \times 10^{-3}$
G_{sinc} (ρ_{44838}, 384-bit)	$h = \frac{5}{8}$	89676.89867887534 − 0.0105638607534751 i	10^{-30}
G_{sinc} (ρ_{44838}, 384-bit)	$h = \frac{625121}{10^6}$	89676.89866265337 − 0.0005506986593098 i	10^{-30}
G_{sinc} ($\rho_{2442}, \rho_{44838}$)	$h_* = \sigma^* - \frac{1}{2}$	two disjoint off-real quartets ($N_{\mathrm{off}} \ge 8$)	10^{-27}
H_t^{DH} (ρ_{240}, 512-bit)	$t = \frac{28825920237}{125000000000}$	$480.6902310425075 - 9.535301365553739 \times 10^{-7}$ i	10^{-32}

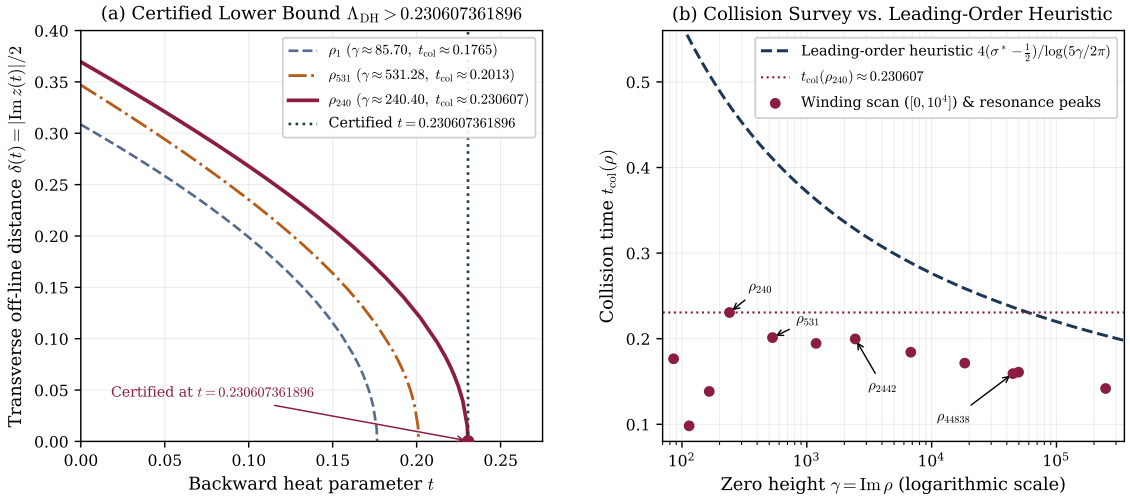


FIGURE 2. (a) Transverse off-line distance $\delta(t) = |\,\mathrm{Im}\, z(t)|/2$ under backward heat flow for ρ_1, ρ_{531}, and ρ_{240}, with the certified 512-bit Interval–Newton witness at $t = 28825920237/125000000000 = 0.230607361896$. (b) Winding scan ($[0, 10^4]$) and high-resonance collision survey compared with the leading-order heuristic $4(\sigma^* - \frac{1}{2})/\log(\frac{5\gamma}{2\pi})$.

and the Argument Principle, $H_{t_*}^{\mathrm{DH}}$ has a unique simple zero in B, which is strictly non-real. If $H_{t_0}^{\mathrm{DH}}$ had only real zeros for some $t_0 \le t_*$, Lemma 4.1(i) applied to $H_{t_*}^{\mathrm{DH}} = \mathrm{e}^{-(t_*-t_0)\partial_z^2} H_{t_0}^{\mathrm{DH}}$ would force all zeros at $t = t_*$ to be real, a contradiction; openness of simple non-real zeros in t yields $\Lambda_{\mathrm{DH}} > t_*$. □

Remark 5.3. Combining Theorem 5.2 with Platt and Trudgian's unconditional upper bound $\Lambda_\zeta \le \frac{1}{5} = 0.2$ [21] separates the two constants strictly under the common normalization of Section 1.2: $\Lambda_\zeta \le \frac{1}{5} < \frac{28825920237}{125000000000} < \Lambda_{\mathrm{DH}}$.

5.2. **Numerical collision survey versus certified theorems.** We strictly distinguish the ball-certified theorems in Table 1 from exploratory numerical root tracking:

(1) *Oscillation aliasing in non-interval quadrature:* Near the first off-line zero $\rho_1 \approx 0.808517 + 85.699348\,\mathrm{i}$ ($z \approx 171.40 - 0.6170\,\mathrm{i}$), $\cos(zu)$ undergoes roughly 60 oscillations on $[0, 2.25]$; uncertified adaptive quadrature (`mpmath.quad` at default subdivision) aliases across sign changes and reports a spurious coalescence near $t \approx 0.1419$, whereas high-precision scalar Newton refinement places coalescence at $t_{\mathrm{col}}(\rho_1) \approx 0.17650538$.

(2) *Exhaustive* $[0, 10{,}000]$ *winding scan and Diophantine resonance survey up to* 10^8*:* High-precision scalar Newton refinement on $(H_t^{\mathrm{DH}}, \partial_z H_t^{\mathrm{DH}}) = (0, 0)$ locates the coalescence of ρ_{240} at $t_{\mathrm{col}}(\rho_{240}) \approx 0.2306073618964546$ ($x_{\mathrm{col}} \approx 480.6902310425$, matching the first 12

decimal digits of Theorem 5.2) and that of ρ_{531} at $t_{\mathrm{col}}(\rho_{531}) \approx 0.20127583$. An argument-principle winding scan over $\gamma \in [0, 10{,}000]$ (572 off-line zeros in $\operatorname{Re} s > \frac{1}{2}$) combined with a Kronecker phase-alignment search up to $\gamma = 10^8$ (Figure 2(b)) detects no collision exceeding $t_{\mathrm{col}}(\rho_{240})$: although deeper real parts occur at higher ordinates (such as ρ_{44838} with $\operatorname{Re}\rho \approx 0.91614$ and $t_{\mathrm{col}} \approx 0.1592$, or $\gamma \approx 247{,}535.1$ with $\operatorname{Re}\rho \approx 0.94612$ and $t_{\mathrm{col}} \approx 0.1418$), matching the Gaussian saddle $u_0 \approx \frac{t}{4}\log\frac{5\gamma}{2\pi}$ to $\sigma^* - \frac{1}{2}$ suggests the unproved leading-order heuristic $t_{\mathrm{col}}(\rho) \lesssim 4(\sigma^* - \frac{1}{2})/\log(\frac{5\gamma}{2\pi})$, which itself falls below $t_{\mathrm{col}}(\rho_{240})$ for all $\gamma > 59{,}220$ (whereas Proposition 4.3 gives only the much weaker rigorous ceiling $4(\sigma^* - \frac{1}{2} + 5 + 6\sqrt{t} + 2t)/\log(\frac{5\gamma}{2\pi})$). Whether Λ_{DH} coincides with $t_{\mathrm{col}}(\rho_{240})$ remains an open finite-box verification problem.

6. Scope, Limitations, and Open Questions

Theorem 1.1 shows that for self-dual Dirichlet combinations without an Euler product, real entire Pólya–Schur multipliers exhibit both an infinite-height shift threshold at $S_G(u) \to 2\sigma^* - 1$ (sensitive to root-gap regularity even when $\rho(G) \in (1/2, 1)$) and a strict gap between infinite-height real-rootedness and global real-rootedness for strip-averaging kernels such as $G_{\mathrm{sinc},h}$, bracketed within a factor of two by $\sigma^* - \frac{1}{2} < \frac{5}{8} < h^*_{\mathrm{global}}(G_{\mathrm{sinc}}; \Xi_{\mathrm{DH}}) \le 2\sigma^* - 1$. Alongside removing RH and GRH in Proposition 3.3 and resolving the remaining oscillatory and clustered boundary regimes of Remark 2.8, closing the finite-height gaps $\frac{5}{8} < h^*_{\mathrm{global}}(G_{\mathrm{sinc}}; \Xi_{\mathrm{DH}}) \le 2\sigma^* - 1$ and $\frac{28825920237}{125000000000} < \Lambda_{\mathrm{DH}} < 2(\sigma^* - \frac{1}{2})^2$ reduces to finite two-dimensional argument-principle covers once the height cutoffs are sharpened beyond the ℓ^1 barrier of Remark 4.5.

Appendix A. Proofs of Explicit Analytic Constants and Gaussian Tail Bounds

Proof of Lemma 4.2. Set $S(N) := \sum_{n=1}^{N} a_{\mathrm{DH}}(n)$. Because $a_{\mathrm{DH}}(n+5) = a_{\mathrm{DH}}(n)$ and $\sum_{r=1}^{5} a_{\mathrm{DH}}(r) = 1+\kappa-\kappa-1+0 = 0$, checking $N \equiv 1, 2, 3, 4, 0 \pmod 5$ gives $\max_{N\ge1} |S(N)| = 1 + \kappa < 1.2841$. By conjugation symmetry $\mathcal{D}_{\mathrm{DH}}(\sigma - \mathrm{i}\tau) = \overline{\mathcal{D}_{\mathrm{DH}}(\sigma + \mathrm{i}\tau)}$, assume $\tau \ge 0$, set $\sigma := \frac{1}{2} + v \ge \frac{1}{2}$, and first consider $\tau \ge 100$ with $M := \lfloor \tau \rfloor \ge 100$. Splitting $\mathcal{D}_{\mathrm{DH}}(\sigma + \mathrm{i}\tau) = \sum_{n=1}^{M} a_{\mathrm{DH}}(n) n^{-\sigma-\mathrm{i}\tau} + \sum_{n=M+1}^{\infty} a_{\mathrm{DH}}(n) n^{-\sigma-\mathrm{i}\tau}$, the prefix satisfies

$$\begin{aligned}\left|\sum_{n=1}^{M} a_{\mathrm{DH}}(n) n^{-\sigma-\mathrm{i}\tau}\right| &\le 2(1+\kappa) \sum_{m=0}^{\lfloor (M-1)/5 \rfloor} (5m+1)^{-1/2} \\ &\le 2(1+\kappa)\Big(1 + \frac{2}{5}(\sqrt{\tau} - 1)\Big) \le 2(1+\kappa)(0.46\tau^{1/2}) < 1.19\,\tau^{1/2}.\end{aligned} \tag{A.1}$$

Summing $n \ge M+1$ by parts yields $-S(M)(M+1)^{-\sigma-\mathrm{i}\tau} + (\sigma + \mathrm{i}\tau)\int_{M+1}^{\infty} S(\lfloor x \rfloor) x^{-\sigma-1-\mathrm{i}\tau}\,\mathrm{d}x$. For $\sigma \le 19\tau^{1/2}/20$, $|\sigma + \mathrm{i}\tau| \le 1.005\tau$ and $\int_{M+1}^{\infty} x^{-3/2}\,\mathrm{d}x < 2\tau^{-1/2}$, bounding the tail by $(1+\kappa)\tau^{-1/2} + 2.01(1+\kappa)\tau^{1/2} < 2.71\tau^{1/2}$ and $|\mathcal{D}_{\mathrm{DH}}| < 3.90\tau^{1/2} < 4\tau^{1/2}$; for $\sigma > 19\tau^{1/2}/20 \ge \frac{19}{2}$, $|\mathcal{D}_{\mathrm{DH}}| \le \zeta(19/2) < 1.002 < 4\tau^{1/2}$. When $0 \le \tau \le 100$, the same split at $M = 100$ gives $< 11.82 + 0.13 + 25.80 < 38$ for $\sigma \le \frac{19}{2}$, and $\zeta(19/2) < 38$ for $\sigma > \frac{19}{2}$. Finally, for $u \ge \frac{5}{4}$ ($\sigma \ge \frac{7}{4}$), Hurwitz evaluation at $\sigma = \frac{7}{4}$ gives $\sum_{n=2}^{\infty} |a_{\mathrm{DH}}(n)| n^{-7/4} = 0.41770137\ldots < \frac{1}{2}$, whence $\operatorname{Re}\mathcal{D}_{\mathrm{DH}} > \frac{1}{2}$ and $|\mathcal{D}_{\mathrm{DH}}| < \frac{3}{2}$. □

Completion of Binet and tail bounds in Proposition 4.3. Write $z_0 := \frac{3}{4} + \frac{\mathrm{i}\tilde{\gamma}}{2}$, so $|z_0| > \frac{\tilde{\gamma}}{2} \ge 200$. In Binet's second integral $\psi(z) - \log z + \frac{1}{2z} = -\int_0^{\infty} \frac{2s\,\mathrm{d}s}{(s^2+z^2)(\mathrm{e}^{2\pi s}-1)}$ at $z = z_0 + w$ ($w \ge 0$), splitting at $s = |z|/\sqrt{2}$ (where $|s^2 + z^2| \ge \frac{1}{2}|z|^2$ on $[0, |z|/\sqrt{2}]$ and $|s^2 + z^2| \ge 2(\operatorname{Re} z)(\operatorname{Im} z) \ge 300$ on $(|z|/\sqrt{2}, \infty)$) yields $|\psi(z_0 + w) - \log(z_0 + w) + \frac{1}{2(z_0+w)}| \le \frac{1}{5|z_0|^2}$, and integrating over $w \in [0, u/2]$ together with $|\log(1 + \frac{w}{z_0})| \le \frac{w}{|z_0|}$ and $|\log(1 - \frac{3\mathrm{i}}{2\tilde{\gamma}})| \le \frac{3}{2\tilde{\gamma}}$ gives $|E_{\mathrm{St}}(u, \tilde{\gamma})| \le \frac{u^2+6u}{4\tilde{\gamma}}$ for all $u \ge 0$. Set $X := \log\frac{5\tilde{\gamma}}{2\pi}$. By (4.2), $\frac{t}{4}X > \Delta + 5 + 6\sqrt{t} + 2t$

and $\tilde\gamma > \max(64t^3, \mathrm{e}^X) \ge 16t^2\mathrm{e}^{X/3}$. Since $u_0 + 3\sqrt{t} \in (5, \frac{t}{2}X)$, on $u \in [0, u_0 + 3\sqrt{t}]$ we have $|E_{\mathrm{St}}(u,\tilde\gamma)| < \frac{11}{1280}X^2\mathrm{e}^{-X/3} < 0.042 < \frac{1}{10}$. On $u \ge u_0 + 3\sqrt{t}$, $\frac{u^2+6u}{4\tilde\gamma} < \frac{1}{10} + \frac{(u-u_0)^2}{2t}$, so Lemma 4.2 bounds the right tail by $\frac{3}{2}\mathrm{e}^{1/10}\int_{3\sqrt{t}}^{\infty}\mathrm{e}^{-y^2/(2t)}\,\mathrm{d}y < 0.01\sqrt{\pi t}$. On $[\frac{5}{4}, u_0 - 3\sqrt{t}]$, Lemma 4.2 bounds the integral by $\frac{3}{2}\mathrm{e}^{1/10}\int_{3\sqrt{t}}^{\infty}\mathrm{e}^{-y^2/t}\,\mathrm{d}y < 10^{-4}\sqrt{\pi t}$. On $[0, \frac{5}{4}]$ and $\mathcal{V}_\theta$, $\frac{(u_0-5/4)^2}{t} > \frac{17}{32}X + \frac{2.1}{t} + \frac{6.75}{\sqrt{t}} + 5.4 + 0.6t$ and $|\mathcal{D}_{\mathrm{DH}}| \le 4\tilde\gamma^{1/2} < 4.52\mathrm{e}^{X/2}$ bound the $[0, \frac{5}{4}]$ piece by $< 10^{-3}\sqrt{\pi t}$, $|\mathcal{E}_{\mathcal{V}}| < 10^{-3}\sqrt{\pi t}$, and $|\xi_{\mathrm{DH}}(\frac{1}{2} + \mathrm{i}\gamma)/(\sqrt{\pi t}\,A_t(\gamma,0))| < 10^{-2}$. Finally, integrating $\frac{2|u-u_0|}{t}\mathrm{e}^{-(u-u_0)^2/t}$ over $[\frac{5}{4}, u_0 + 3\sqrt{t}]$ contributes $< \frac{1.88}{\sqrt{t}}\sqrt{\pi t}$ (plus $< \frac{0.06}{\sqrt{t}}\sqrt{\pi t}$ from the tails), and dividing by $0.29\sqrt{\pi t}$ gives $< \frac{6.7}{\sqrt{t}} < \frac{1}{4}X$, proving (4.4) and (4.5). □

Appendix B. Ball-Arithmetic Truncation Enclosures

Proof of Lemma 5.1 and Cauchy bound in Theorem 3.2. For $n \ge N \ge 25$ and $u \in [0, U]$, $\mathrm{e}^{4u} \ge 1$ and $n^2 \ge Nn$ give $\mathrm{e}^{-\pi n^2\mathrm{e}^{4u}/5} \le r_N^n$ with $r_N = \mathrm{e}^{-\pi N/5} < 10^{-6}$. Since $|a_{\mathrm{DH}}(n)| \le 1$, summing $\sum_{n=N}^{\infty} n\, r_N^n = \frac{r_N^N(N-(N-1)r_N)}{(1-r_N)^2}$ and using $|\partial_z^k \cos(zu)| \le U^k\mathrm{e}^{|y|U}$ over $[0,U]$ gives $E_1(U,N,k,|y|)$. For $u \ge U \ge 2$ and $k \in \{0,1,2,3,4\}$, $\sum_{n=1}^{\infty} n\,\mathrm{e}^{-\pi n^2\mathrm{e}^{4u}/5} \le 2\mathrm{e}^{-\pi\mathrm{e}^{4u}/5}$ and $tu^2 + (3+|y|)u + k\log u \le \frac{u^2}{4} + 8u \le \frac{\pi}{10}\mathrm{e}^{4u} - 4u$ $(u \ge 2)$, so the tail on $[U,\infty)$ is bounded by $16\mathrm{e}^{-4u-\frac{\pi}{10}\mathrm{e}^{4u}} \le \frac{40}{\pi}\mathrm{e}^{-4U}\cdot\frac{2\pi}{5}\mathrm{e}^{4u}\mathrm{e}^{-\frac{\pi}{10}\mathrm{e}^{4u}}$, integrating to $E_2(U)$. Bounding $\sum_{n=1}^{N-1} n|a_{\mathrm{DH}}(n)|\,\mathrm{e}^{-\pi n^2\mathrm{e}^{4u}/5} \le \frac{N(N-1)}{2}$ and $|u^4\cos(zu)| \le U^4\cosh(Uy_{\max})$ on $[0,U]$ yields (5.2). Finally, in Theorem 3.2(i), $\tilde\xi_{\mathrm{DH}}(s_{\mathrm{ref}} + w)$ is entire in w; covering $|w| = 3$ around $s_{\mathrm{ref}} = \frac{1}{2} + 44838.46\,\mathrm{i}$ by 256 overlapping interval arcs $W_j := 3\mathrm{e}^{\mathrm{i}\Theta_j}$ $(\Theta_j = \frac{2\pi}{256}[(j+\frac{1}{2}) \pm 0.5001])$ in 384-bit `FLINT/Arb` certifies $M_3 := \max_{0\le j<256}\sup_{w\in W_j}|\tilde\xi_{\mathrm{DH}}(s_{\mathrm{ref}} + w)| \le 5.934 \times 10^8$, whence Cauchy's inequality $|c_k| \le M_3 \cdot 3^{-k}$ on $|s(B_k) - s_{\mathrm{ref}}| + \sup|H_k| < \frac{6}{5}$ bounds all Taylor remainders in (3.5) and its z- and h-derivatives (and similarly via 128 arcs on $|w| = 2$ around $0.71 + 2442.49\,\mathrm{i}$ for the ρ_{2442} enclosure in Proposition 3.3). □

Use of AI and computational verification

Portions of the literature exploration, asymptotic stress-testing, manuscript drafting, and `FLINT/Arb` verification scripting were conducted with the assistance of an AI research assistant (**Gemini**). Every analytical inequality, contour deformation, Möbius inversion identity, and interval-arithmetic enclosure reported in this paper was independently checked against symbolic and arbitrary-precision ball-arithmetic computations; all proof certificates are deterministically verified by the open-source script `scripts/verify_paper15_debruijn_newman_dh.py` using `python-flint` [12] (with optional figure regeneration via `--regenerate-figures`).

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